



Ansökan

2024-09-12

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Medelsförvaltare: Sveriges lantbruksuniversitet					
Hemvist: Akvatiska resurser					
Information om ansökan					
Utlisningsnamn: För en ekosystembaserad akvatisk förvaltning 2024 (Miljöforskningsanslaget)					
Bidragsform: Projekt					
Sökt inriktning: Forskningsprojekt					
Ämnesområde utlysning: Miljöforskningsanslaget					
Projekttitel: Variationer i sälars födoval - Viktig kunskap för utveckling av ekosystembaserad förvaltning					
Projektstart: 2025-03-01			Projektslut: 2027-12-31		
Sökt beredningsgrupp: BG: MPREBF					
Klassificeringskod: 10611. Ekologi					
Sökta medel					
År:	2025	2026	2027	Totalt sökt belopp	
Belopp:	542 121	1 447 349	1 138 604	3 128 074	
Medverkande					
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Hur fick du information om utlysningen?

Lärosätet eller registrator

Om du valt "Annat", vänligen ange här.

Projektinformation

Projekttitel (svenska)

Variationer i sälars födoval - Viktig kunskap för utveckling av ekosystembaserad förvaltning

Projekttitel (engelska)

Long-term feeding ecology of harbour seals: Implication for sustainable ecosystem-based management

Abstract och populärvetenskaplig beskrivning

Sammanfattning (svenska)

Efter framgångsrika bevarandeinsatser har knubbsälarna (*Phoca vitulina*) i Skagerrak återhämtat sig från rekordlåga nivåer under 1970-talet. Precis som i många andra ekosystem där marina däggdjur ökat i antal har konflikter mellan sälar och fiske blossat upp, ofta förstärkta av okunskap om sälarnas påverkan på fisk. För att förvaltare ska kunna tillämpa ekosystembaserade förvaltningsstrategier är det viktigt att kvantifiera sälarnas ekologiska roll, som i nuläget begränsas av bristen på information om sälarnas födoval.

I detta projekt kommer vi att undersöka sälarnas födoval, och hur det varierar rumsligt, över tid och med sälarnas ålder och kön samt i relation till tillgången på fisk. Vi kommer att analysera unika data baserat på över 2000 dietprover från de senaste två decennierna och med statistiska modeller öka förståelsen av interaktioner mellan säl och fisk. Projektet går längre än tidigare studier, som varit begränsade med avseende på 1) geografisk omfattning 2) föråldrade data med låg tidsmässig upplösning och i huvudsak baserade på spillningsprov som inte ger information om sälens, ålder eller kön. Dessutom kommer vi att använda motsvarande långsiktiga data på kustfisk från provfisken. Det ger en unik möjlighet att studera sälarnas födoval i relation till lokal tillgång på bytesfisk och dess storleksstruktur, vilket bidrar med kunskap om i vilken utsträckning sälarna kan anpassa sin diet efter bytestillgång eller om de har specifika preferenser. Dessutom kommer forskningen att utvärdera sälpredationens inverkan på fiskdödligheten i förhållande till andra källor.

Genom att tillhandahålla grundläggande kunskap för en ekosystembaserad förvaltning kommer våra resultat att bidra till att sammanföra och höja kunskapsnivån hos olika intressenter. Insikterna från projektet kommer att stödja faktabaserade och adaptiva ekosystembaserade förvaltningsstrategier för balansering mellan en livskraftig knubbsälspopulation och en hållbar fiskeriförvaltning.

Abstract (Engelska)

In response to conservation efforts, the harbour seal (*Phoca vitulina*) population in the Skagerrak has recovered from the record low abundance of approximately 2,000 individuals in the 1970's following overexploitation. As in many other ecosystems with recovering marine mammal populations, this has led to conflicts between seals and fisheries, often on limited scientific evidence. Key to moving beyond this conflict is to quantify the ecological role of seals, such that managers can balance tradeoffs within an ecosystem-based management approach. However, we currently lack crucial information about harbour seal feeding ecology. In this project we will assess the magnitude and uncertainty of the spatiotemporal dynamics of seal feeding and its relationship to local fish abundance, to better understand seal-fish interactions along the Skagerrak coast.

In this project, we will analyze a unique dataset of over 2,000 diet samples collected over the past two decades. This project moves beyond previous studies on this topic, which are outdated and limited by low spatial and temporal resolution, and based on scat samples, which do not provide information on age or sex of the seals. Importantly, a corresponding long-term coastal fish monitoring dataset provides unique opportunity to study predator diet choices in relation to local prey availability and its size structure, allowing us to determine to what extent seals are opportunistic feeders adapting to dynamic prey abundances. Additionally, the research will evaluate the impact of seal predation relative to other sources of fish mortality.

By providing essential evidence to refine management strategies, our findings will bridge the gap between stakeholder concerns and scientific understanding. The insights derived from this project will support a balanced, evidence-based and adaptive approach to harbour seal conservation and sustainable fisheries management.

Populärvetenskaplig beskrivning (svenska)

Tack vare riktade bevarandeåtgärder har knubbsälspopulationen i Skagerrak återhämtat sig från en mycket låg nivå under 1970-talet, orsakad av omfattande jakt och påverkan från miljögifter. I takt med att antalet sälär ökat har också konflikterna med fisket ökat. Olika former av jakt har därför återinförts för att skydda fisk och fiske, trots att det saknas kunskap om och hur sälarna påverkar olika fiskbestånd.

Sälars roll i ekosystemet och deras effekter på fisk och fiske är flitigt debatterad och konfliktdrabbade ämnen. Grundläggande kunskap om vad knubbsälar äter, hur dieten varierar över året och mellan olika livsstadier och, framför allt, hur sälarna påverkar ekosystemet saknas. Sådan information är nödvändig för att förstå sälarnas ekologi och betydelse i ekosystemet.

Tidigare undersökningar från 1970- och 1980-talen är begränsade till ett fåtal områden och tidpunkter. Framför allt saknas information från vintermånaderna, hur dieten varierar mellan områden och hur dieten beror på sälarnas ålder och kön. Med tanke på de stora förändringarna som skett i havet, inte minst de dramatiskt minskande mängden fisk, behövs uppdaterade och mer omfattande dietstudier för att få en bild av hur kopplingen mellan sälär och tillgången på fisk ser ut. Att förstå hur tillgången på fisk påverkar sälarnas födoval är avgörande, inte bara för att utöka förståelsen om sälarnas ekologi, utan även för att mer korrekt kunna undersöka interaktionerna mellan sälär, fiskar och de fisken som är utre efter samma fiskar.

Vårt projekt erbjuder en unik möjlighet att ta fram ny, nödvändig, kunskap för utveckling av en ekosystembaserad förvaltning. Vi kommer att analysera dietprover som samlats in under de senaste två decennier för att få en bättre insikt i sälarnas konsumtion av fisk. Vi kommer inte bara kunna svara på "vad" sälär äter, utan också "vem" inom sälpopulationen (hanar jämfört med honor eller unga sälär jämfört med vuxna) som äter vad. Kunskap om hur ålder och kön påverkar vad sälarna äter är viktig för att skatta födobehovet på ett bättre sätt.

Vår studie blir den första att undersöka hur sälarnas diet påverkas av tillgången på lokala bytesarter. Genom att kombinera dietdata med provfiskedata kommer vi att kunna ta reda på om sälarna är selektiva i sina val av byten eller om de är opportunistiska och äter de fiskar som är lättast tillgänglig. Med sådan information kan man förutspå hur sälär anpassar sin fiskkonsumtion som svar på förändringar i fisktillgång eller miljöförändringar, och även vilken påverkan sälarna kan ha olika fiskbestånd.

Vi kommer att uppskatta hur mycket sälarna äter av olika arter och storlekar av fisk. Detta gör det möjligt för oss att bedöma sälarnas påverkan på fisk jämfört med andra orsaker, till exempel fiske.

Genom att klargöra vad sälarna äter och hur det varierar kommer vi att bidra med viktig kunskap för bättre förvaltningsstrategier. Att förstå hur sälär, fisk och fiske samspelar är viktigt, inte bara för att upprätthålla livskraftiga sälpopulationer, utan också för att uppnå samexistens mellan sälär, fiskar och mänskliga aktiviteter. På så sätt kan projektet minska konflikter mellan olika intressenter genom att tillhandahålla kunskap om sälarnas matvanor och betydelse i ekosystemet. Det kommer att bidra till utveckling av en ekosystembaserad förvaltning och mer informerade och balanserade beslut för bevarandet av sälär, fisk och fiske.

Projekttid

Sökt antal månader

34

Beräknad projekttid

2025-03-01 - 2027-12-31

Klassificeringar

SCB-koder

10611. Ekologi

Miljökvalitetsmålen

10. Hav i balans samt levande kust och skärgård

Generationsmålet

5. En god hushållning sker med naturresurserna.
2. Den biologiska mångfalden och natur- och kulturmiljön bevaras, främjas och nyttjas hållbart.
1. Ekosystemen har återhämtat sig, eller är på väg att återhämta sig, och deras förmåga att långsiktigt generera ekosystemtjänster är säkrad.

ORCID

Ange ditt ORCID.

0000-0001-7844-6086

Etiska överväganden

I projektet ingår djurförsök

Nej

Redovisning av djurförsök och etiska överväganden

Diet samples from seals have, and will continue to be, collected as scat samples from seal haul outs and digestive tracts from already ongoing hunting. No specific ethical permits are needed for collection and analysis of seal diet samples. Fish monitoring data is extracted from an existing database, originating from a long-term coastal fish survey¹ for which ethical approval has been granted by the Swedish Board of Agriculture. All data collection and use are conducted in accordance with relevant ethical standards and regulations.

Inriktning

Inriktning

B: Praktisk förvaltning av sälarterna och storskarv

A: Ekosystembaserad akvatisk förvaltning i ett systemperspektiv

C: En effektiv, adaptiv och legitim förvaltning

1. Mål och förväntade resultat

Scientific aims and purpose of the project

This project aims to describe the feeding ecology of harbor seals (HS) in the Skagerrak, focusing on their prey choices relative to the species composition and size structure of fish communities. By analyzing long-term diet data alongside local prey abundance from coastal trawl surveys and applying spatiotemporal statistical and bioenergetic models, we will assess whether HS prefer specific prey or feed opportunistically based on availability and determine if HS diet can serve as an indicator of ecosystem changes. We will evaluate the impact of seal predation on fish populations by comparing consumption estimates to other sources of fish mortality² and explore prey selection differences among age groups and sexes. The study will also consider the size structure of prey fish to assess the effect of seal predation on fish recruitment. This approach will improve understanding of how top predators like HS influence fish communities, providing valuable insights for ecosystem-based management (EBM) strategies to support sustainable coexistence of seals, fish, and fisheries.

Background

The role of marine mammals in ecosystems, particularly their impacts on fish stocks and commercial fisheries, has gained renewed attention as some populations recover from culling, overexploitation, and other human impacts³. HS populations in the Skagerrak and Kattegat regions reflect this trend. Once drastically reduced due to overexploitation in the 20th century, with numbers dropping from about 17,000 to 2,000 individuals by the 1970s^{4,5}, these populations have shown signs of recovery due to conservation efforts. However, conflicts between seals and fisheries have increased, leading to the reintroduction of hunting as a management measure in 2009, with regional quotas reaching up to 900 seals annually from 2020 to 2022⁶. Despite recovery efforts, seals remain vulnerable to threats like recurring viral epizootics that have caused significant declines^{5,7,8}. Recent data indicate that while the HS population grew after the 1980s, growth rates have stalled over the past decade, with possible declines raising concerns about long-term viability⁹. Fish stocks that support both commercial fisheries and marine predators have been severely depleted by overfishing, habitat degradation, and climate change^{2,9}. In Skagerrak, the cod stock, shows a negative growth trend due to low recruitment and small stock size¹⁰. As coastal, generalist predators with strong site fidelity, HS primarily forage in shallow waters, feeding on medium-sized prey like cod, sprat, herring, and flatfish¹¹⁻¹⁷. As top predators, they can significantly influence the structure and function of the ecosystem¹⁸.

Current state of knowledge

There is a lack of long-term studies on seal diets along the Swedish coast of Skagerrak. Existing research is geographically and temporally limited and lacks seasonal sampling, often relying on scat samples from single locations to infer dietary patterns^{11-14,17,19}. Most studies date back to the 1970s and 1980s, covering only a few years and seasons¹¹⁻¹⁴. Given the ecological changes since then, updated research is critically needed to better understand seals' feeding behavior. Current studies do not provide insights into age- and sex-specific feeding behavior of HS or account for temporal variations in prey availability.

While scat analysis can identify prey species and sizes, it does not provide demographic information on the predators (e.g., age, sex) or the number of contributing individuals²⁰. Analysis of prey remains from the digestive tracts of dead seals (hunted or bycaught) allows linking prey choice with demographic traits but has been limited to a single study with only 8 samples¹⁶. Thus, there is no information on how prey choice varies by age or sex within the HS population. Studies of other HS populations suggest that prey preferences differ among age groups and sexes²¹⁻²⁵.

Moreover, no research has examined the relationship between HS diet and the local abundance of prey populations. It is unclear whether seals feed proportionally to prey availability or show specific preferences, limiting understanding of their adaptive feeding strategies and impact on local prey populations. There is also insufficient knowledge about HS total prey consumption and resource needs^{17,26}.

Research questions

This project will process a unique time series of diet data and address the following questions:

Q1: How does the diet of HS vary across different regions, seasons, and years?

Q2: How is the diet of HS influenced by the local abundance and distribution of prey species?

Q3: How do age and sex affect the dietary preferences and feeding habits of HS?

Q4: How do the total prey consumption and resource requirements of HS for key species fluctuate seasonally and over time?

Expected Results

Understanding HS diet patterns: The analysis will determine whether HS have consistent dietary patterns or if their prey preferences have changed over time due to shifts in prey availability, using spatiotemporal models. It will also explore how diet varies by age and sex, distinguishing between natural dietary variations and those caused by external pressures like overfishing or climate change.

Consumption rates quantification: By applying bioenergetic models, we will estimate the energy needs and food choices of HS based on the energy content of different fish species, addressing the question, "How much fish does the seal population need?" rather than just "How much fish is removed?" This approach will assess uncertainties related to population size, diet composition, and bioenergetic parameters, providing a clearer picture of consumption rates.

Support for EBM: The findings will clarify the role of HS in the marine food web, informing policies that balance seal conservation with sustainable fishing practices, in line with EBM principles that aim for the health of entire ecosystems rather than individual species.

Figur

2. Metoder

The project is structured into three work packages:

WP1 Sample Processing and Diet Analysis

Long-term datasets are invaluable for detecting trends and shifts in ecological processes that may not be apparent in short-term studies, especially given the high individual variation in diet data. For HS, long-term data on diet composition can reveal how their feeding behavior has changed in response to various environmental pressures, including fluctuations in prey availability, as well as increasing seal numbers. Such data can also provide insights into the resilience of seal populations to ecological disturbances and their ability to adapt to changing environmental conditions²⁷.

We propose to process and analyze a rare and extensive collection of over 2000 diet samples from HS, gathered along the Swedish coast of Skagerrak over the past 25 years (spanning from 1999 to 2024; Fig. 1). This comprehensive dataset includes scats, collected at haul outs, and digestive tracts, collected from previous and

ongoing hunting programs, providing a unique source of long time-series of data for understanding seal feeding ecology. In Sweden, HS have been hunted since 2009 and stomachs and intestines have been collected by the Swedish Museum of Natural History and the Swedish University of Agricultural Sciences. Diet samples from hunted seals are combined with background information and tissue samples, providing a valuable opportunity to link prey choice with ontogenetic details, such as the age and sex of individual seals.

Diet determination in seals relies on the morphological identification of prey items in diet samples. Prey remains will be determined to lowest possible taxa, i.e., species or genus/family, by examination of otoliths, but also other skeletal parts and whole fish found in the samples. Otoliths are calcium carbonate structures in the inner ear of vertebrates. The number of sagittae otoliths can be used to infer a minimum number of prey individuals ingested for a given fish species. Moreover, as the size of otoliths correlate with the length and weight of a fish, detailed measurements of otoliths can be used to estimate prey size and biomass. This information has large implications for quantifications of seal-fishery interactions and the energetic importance of the diet.

While otoliths and other hard parts from scats and digestive tracts are commonly used to infer seal diets, these methods have some limitations that we aim to address in our study. Firstly, the data derived from otoliths and hard parts typically reflect the diet of seals within a few hours or days prior to sampling, depending on the gut passage time of the prey. This means that the prey identified often represents only those consumed in close proximity to the collection site and during the sampling period²⁸. However, this limitation is mitigated in our study by our samples span multiple occasions throughout the year, capturing seasonal variations and providing a broader temporal context. Secondly, otoliths can be partially or fully digested, making it challenging to identify prey to the species level or even to detect certain species altogether^{29,30}. To overcome this challenge, we will utilize a novel machine learning model developed by our team³¹, which is specifically designed to distinguish between different gadoids, a group of species often central to seal-fishery conflicts and typically difficult to differentiate using traditional methods. By integrating these advanced techniques, our study will provide a more accurate and detailed understanding of HS diets, addressing some of the limitations inherent in conventional otolith-based analyses.

WP2 Prey Selectivity and Spatial/Temporal Analysis

Data sampled over space and time (spatiotemporal) are rich in ecological information, but also case spatially correlated residuals, which if not accounted for can lead to overconfidence in parameter estimates. Covariates, such as temperature or depth, can often explain some of these spatially structured patterns, but not all. In this project, we will use a mixed modelling approach to understand the spatiotemporal variation in seal feeding. Specifically, we will fit generalized linear mixed models (GLMMs) to georeferenced diet data, and model spatial structure using Gaussian random fields (GRFs). GRFs are computationally demanding to fit, so we will leverage the stochastic partial differential equation (SPDE) approach to approximate GRFs with Gaussian *Markov* random fields (GMRFs). This is an increasingly popular approach to spatial modelling due to the speed and flexibility of fitting these with the Template Model Builder (TMB)³² software and the R-package INLA³³ to set up sparse matrix structures required for the SPDE approach, all of which can be interfaced with the R-package sdmTMB³⁴.

We will fit such spatiotemporal GLMMs to data prepared in WP1 to better understand the spatiotemporal dynamics of HS feeding. The total weight of a prey species in a single seal stomach constitutes an observation (row) and will be modelled using either a hurdle/delta model (binomial + gamma/lognormal models) or a Tweedie model to account for the fact that such diet observations are positive continuous and contain zeroes. This approach has been shown to fit stomach content data well in Atlantic cod in studies by co-applicant M. Lindmark^{35,36}. We will include ecologically reasonable covariates, such as day-of-the year, environmental covariates, distance to shore, as well as spatial and spatiotemporal random effects (modelled as GMRFs). With these models, we will be able to quantify 1) how the prey composition of seals varies over space and time, 2) how it varies seasonally and 3) how the diet composition co-varies with the prey availability in the environment. The latter will be quantified with similar spatiotemporal GLMMs fitted to another unique corresponding long-term dataset of coastal fish monitoring data collected along the same coastline¹ (Fig. 1). This coastal survey has been done annually since 2001 with the purpose of monitoring species composition and recruitment in the benthic fish community in general with particular emphasis on monitoring presence of large cod (*Gadus morhua*).

WP3 Consumption modelling

Total consumption can rarely be estimated precisely. Our aim here is to develop a transparent framework that propagates parameter uncertainty when estimating consumption and uncertainty intervals. This will reveal which sources contribute to uncertainty in total consumption, which will guide future research areas.

In principle, total consumption of a specific prey is given by the product of seal abundance, the average

proportion of that prey in the seal diet in terms weight, and the average total ingestion rate of the seal. The average total ingestion rate in turn stems from estimates of the energetic needs of a seal and the energy density of the prey. A more detailed description of these calculations can be found in Box 1.

This approach will provide annual estimates of total consumption by seals, along with associated uncertainties, on local fish stocks, allowing us to compare these estimates with fishery extractions. The results will offer critical insights for the management of local, depleted fish stocks, helping to inform EBM strategies that balance seal populations with sustainable fishery practices.

Figur

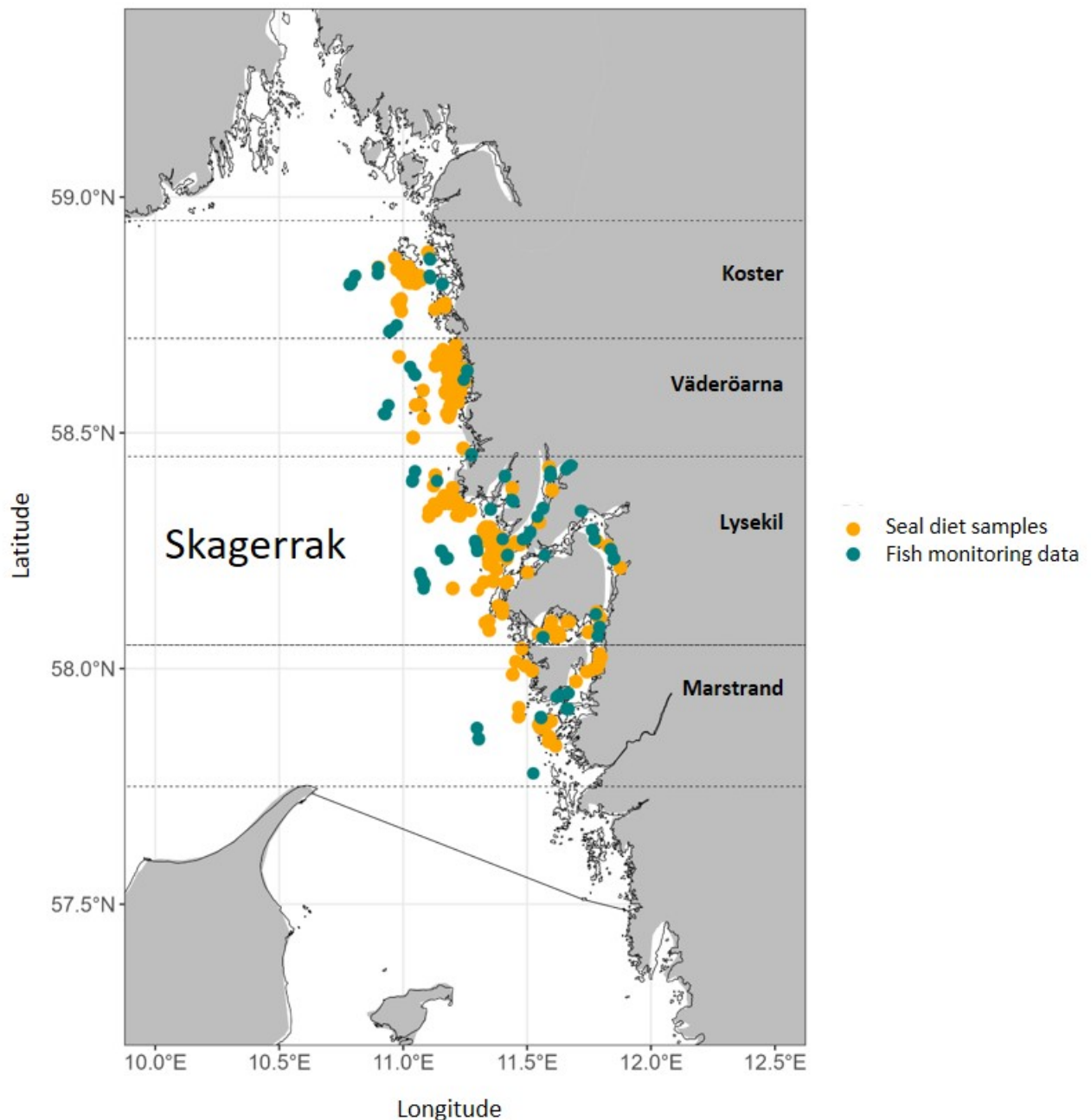


Figure 1. Spatial distribution of seal diet samples (in orange) and fish monitoring data (in green) collected between 1999 and 2024 along the Skagerrak coast. Dashed lines indicate specific local harbour seal sub-populations.

Box 1: Calculating total annual seal consumption by prey species

Total annual consumption $C_{y,s,i}$ by seals in year y on prey species s for iteration i ($i = 0, \dots, 1000$) is the product of total seal abundance $N_{y,i}$, the proportion of the prey in seal stomachs $\%W_{y,s,i}$ and the average total consumption rate $TC_{y,s,i}$ in units $t/year$, i.e., $C_{y,s,i} = N_{y,i} \times \%W_{y,s,i} \times TC_{y,s,i}$. For each iteration i , we sample a single draw from probability distribution describing probably parameter values; a diet composition by simulating from the joint parameter precision matrix of the spatiotemporal model; and sample a value of seal abundance from a distribution describing possible values. The point estimate of total consumption can be described by the median of these n (e.g., 1000) consumption estimates, and uncertainty intervals are defined as some quantiles of the distribution of consumption values.

Prey proportions are expressed in energy by using prey-specific energy density values. $N_{y,i}$ by age class will be derived from updated population models (matrix models) of the seal population provided by co-applicant K. Harding, and $\%W_{y,s,i}$ is quantified in WP2 using spatiotemporal statistical models. The total consumption $TC_{y,s,i}$ in a year is calculated as the product of the daily consumption rate and the number of days seals consume prey s : $TC_{y,s,i} = DC_{y,s,i} \times DAY_{y,s,i}$, where the seasonal consumption dynamics are quantified in WP2. Daily consumption rates $DC_{y,s,i}$ can be calculated in numerous ways³⁷. We will calculate it by dividing the daily energy expenditure $DEE_{y,i}$ by the product of energy density of the diet $ED_{y,i}$, which is the sum of prey-specific weights and prey-specific energy densities) and the assimilation efficiency ($AE_{y,i}$, reflecting that the entire prey is not always consumed or regurgitated). Daily energy expenditure is given by the product of metabolic rate, scaling allometrically as aM^b , an activity factor (AF , active metabolic rate is higher than resting), and the growth premium (GP , higher energetic costs early in life)³⁸, divided by the assimilation efficiency (ME , proportion of energy available to maintenance and growth).

3. Praktisk relevans

A thorough understanding of species interactions is necessary for the development of an EBM approach. Since the 20th century, marine ecosystems have experienced substantial changes, including the recovery of seal populations, depletion of fish stocks and climate change. Recent stagnation in the growth of the harbour seal populations in Sweden signals the need for up-dated and knowledge-based management strategies.

There is ongoing debate among stakeholders about the impact of seals on fish. However, the lack of comprehensive data has led to disagreements among scientists, managers, stakeholders, policymakers, and the public. Despite uncertainties, there is increasing pressure to reduce seal numbers through hunting aimed at preserving fish stocks. This project aims to clarify the interactions between harbor seals and fish in Skagerrak by providing evidence to assess the selectivity of seal predation and its impact on fish.

The project directly addresses the call for knowledge on prey choice in relation to species and size composition of fish communities, as well as energy requirements and prey consumption of seals. The findings will support developing evidence-based EBM strategies that are scientifically robust and ethically sound, aligning with sustainable marine biodiversity conservation's objectives.

Support for key stakeholders

- **Swedish Environmental Protection Agency (SwEPA) and Swedish Agency for Marine and Water Management (SwAM):** Scientific support needed to develop and update conservation and management (including the national management plan). Advanced models developed through the project will support EBM, ensuring that policies are informed by comprehensive and accurate data. The need for long-term studies of HS prey choice and the correlation between diet and changes in fish stocks, as emphasized in the National Management Plan for Harbour Seals (Havs- och vattenmyndigheten, 2014), is directly addressed through this research. Additionally, the project aligns with the governmental proposition '*Ett levande hav – ökat skydd, minskad övergödning och ett hållbart fiske*' (Prop. 2023/24:156) and contributes to achieving the national environmental objectives: '*A balanced marine environment, flourishing coastal areas and archipelagos*' and '*A rich diversity of plant and animal life*'.
- **County Administrative Boards (CABs):** Detailed knowledge on seal-fish interaction will enhance decision-making and local conservation and management strategies.
- **Fishing Industry:** Clarification of how prey choice varies with fish abundance and quantification of fish consumption by seals. This addresses concerns about negative effects of seals and help balance between seal conservation and sustainable fishing practices.
- **General Public and Policymakers:** Transparent, evidence-based information will make people aware of the nature of seal-fish interactions and foster informed decision-making and policy development, promoting dialogues and balanced marine resource management.

Overall impact

- **Update management strategies:** By offering current data on seal-fish interactions, we will support the development of adaptive EBM strategies, including targeted hunting practices.
- **Inform ecological models:** Findings will lay the groundwork for comprehensive ecological models of the ecosystem, e.g., to evaluate management measures.
- **Resolve stakeholder conflicts:** The outcomes can reconcile differing views among stakeholders, such as fishers, conservationists, policymakers, and the public. The project will foster cooperative management and contribute to developing balanced strategies that align ecological sustainability with the interests of diverse stakeholder groups.

Summary

The proposed project provides a scientific foundation for EBM, supporting sustainable coexistence between HS, fish, and human activities. By offering detailed insights into seal feeding ecology and its ecological implications, the research will inform decision-making and promote balanced management strategies.

Figur

4. Kommunikation

Management agencies will receive information from the project continuously in the regular monthly meetings that occur between SLU and the agencies. More specifically, outcomes from the project will be communicated in relation to consultation meetings arranged by SwEPA for updates on the current state of knowledge before decisions on seal hunting programs. The project will also be communicated to national (SwAM) and local (CAB) management agencies together with the fishery sector in connection with annual meetings. Information will be disseminated to the international organizations ICES and HELCOM by team members that are national expert representatives.

Public outreach will be conducted through a popular science article and press releases together with a dedicated "Scandinavian seals" web page (already under construction) and social media platforms. The SLU communication office will be engaged to ensure dissemination of results to many media outlets.

Scientific outreach will be achieved through publication of peer-reviewed articles and attendance at scientific conferences and workshops.

Figur

Bifoga din kommunikationsplan i enlighet med mallen och instruktionerna. Motiveringen anger du ovan under 4. Kommunikation.

Instruktioner och mallar hittar du på Naturvårdsverkets [hemsida](#).

Spara ifylld mall som pdf, därefter klicka på mappknappen och leta upp pdf-filen på din dator för att ladda upp filen till ansökningsformuläret.

Se nästa sida för bilaga.

Communication plan for projects

Title of the call: Towards ecosystem-based aquatic management - Focus area B — Practical management of the seal species and great cormorant

Title of the project proposal:

Personnel involved - insert lines as needed

Project leader Monica Mion (SLU)

Co-worker 1 Max Lindmark (SLU)

Co-worker 2 Karl Lundström (SLU)

Co-worker 4 Karin Hårding (GU)

Co-worker 5 Diet analyst (SLU)

Communication plan, Projects

	Year 1				Year 2				Year 3			
	1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24	25-27	28-30	31-33	34-36
Communication activities												
<i>Specify all communication activities - insert lines as needed</i>												
Continuous communication and management support to national authorities (The Swedish Environmental Protection Agency, the Swedish Agency for Marine and Water Management, Regional County Administrative Boards, the Swedish Board of Agriculture)	X	X	X	X	X	X	X	X	X	X	X	X
Conferences and presentations												
<i>Specify conferences and other presentations - insert lines as needed</i>												
Conference on the Biology of Marine Mammals								X				
Workshops/seminars/courses												
<i>Specify workshops, seminars, and courses to be organized - insert lines as needed</i>												
ICES Working Group on Marine Mammal Ecology (WGMME)					X				X			
Internal seal-fish WS at SLU					X				X			X
WS with the Swedish Museum of Natural History to relate outcomes of the project to the museum staff's work on monitoring of populations development and health status of the Skagerrak HS population								X				X
WS with institutes and stakeholders on the topic of HS and fish in the Skagerrak (The Swedish Agency for Marine and Water Management, Regional County Administrative Boards, fish biologists, fisheries organisations and politicians)						X				X		
Publications and outreach (reports, guidelines, handbooks, policy briefs, newsletters)												
<i>Specify reporting activities - insert lines as needed</i>												
Popular science report about seal-fish interactions in the Skagerrak, e.g. in "Havsutsikt"											X	
Scientific publications (articles, literature reviews)												
<i>Specify scientific publications - insert lines as needed</i>												
1. Spatiotemporal dynamics of Skagerrak harbour seal feeding in relation to age and sex							X	X	X	X		
2. Skagerrak harbour seal feeding choices in relation to local prey availability							X	X	X	X		
3. Harbor Seal Consumption in the Skagerrak: Implications for Local Fish Stock Management and Ecosystem-Based Strategies									X	X	X	X
Other dissemination (web pages, social media, press releases, citizen science, podcasts, films)												
<i>Specify other dissemination activities - insert lines as needed</i>												
Further development of existing web pages about the project and on the topic of seal-fish interactions	X				X				X			X
Social media about the project and on the topic of seal-fish interactions	X				X				X			X

5. Organisation och budget

All participants (Monica Mion (MM), Max Lindmark (ML), Karl Lundström (KL) and Karin Hårding (KH)) will contribute to the project in their specific areas of competence and collaboratively advance to interpretation and communication of results throughout participation in the planned workshops with stakeholders, at scientific conferences and via peer-reviewed publications. MM will have the overall coordination of the project. The whole project team is ready to interact with a Reference Group appointed by the Swedish Environmental Protection Agency and the Swedish Agency for Marine and Water Management at the start of the project.

The total budget of the project is 3,128,074 SEK including direct and indirect costs for salaries, consultant, and travel costs. The funding requested is divided approximately as follows:

- Diet analyses (28%) processing of scats and digestive traits samples and combining all the data in a unique comprehensive dataset including background information (e.g., date of collection, GPS location, age, sex, where possible).
- Statistical analyses and modelling 67%
- Communication (e.g., workshops, conferences) 5%

WPs' share of resources for analyses is approximately:

- WP1 **Sample Processing and Diet Analysis** 40%
- WP2 **Prey Selectivity and Spatial/Temporal Analysis** 50%
- WP3 **Consumption modelling** 30%

Salaries: A total of **57% (1,775,632 SEK)** of the project budget will be allocated to salary costs, covering the following team members:

- **MM** – 50% time allocated to coordinate the project and lead Work Packages 1, 2 and 3 (WP1-3).
- **ML** – 10% time allocated to contribute to the analyses in WP2-3.
- **KL** – 15% allocation to support analyses in WP1-3 and lead stakeholder interactions. KL will also increase the time dedicated to the project by participating to annual meetings regularly attended with external funding (among others the Marine mammal Society Biennial Conference and the ICES working group on marine mammal ecology [WGMME] to present the project results.
- **KH** – 3% in-kind time allocated to support analyses in WP2-3.
- **Diet analyst** – 38% time allocated to process diet samples.

The total funding requested for salaries, including social security costs, amounts to **1,775,632 SEK**. An additional 150,000 SEK is requested to support collaboration with Karin Hårding's laboratory group. This funding will be used for analyses to estimate the age distribution of the Skagerrak HS population over time, which is crucial for assessing seal consumption in WP3

Indirect costs: 1,142,442 SEK (based on a 54.54% overhead applied by SLU).

Conferences: We plan to present the results of this project at international conferences, including the Biennial Conference on the Biology of Marine Mammals. The allocated 60,000 SEK will cover conference fees and travel expenses.

Publications: we plan at least 3 scientific publications in peer-reviewed open-access scientific journals. SLU already has agreements in place with several publishers that allow publication of research articles OA free of charge. Many of the journals under these agreements are suitable outlets for publications of the project results (e.g. ICES Journal of Marine Science, Fisheries Research, Ambio, Marine Mammal Science, and many others).

Figur

Bifoga din tidsplan i enlighet med mallen och instruktionerna. Motiveringen anger du ovan under 5. Organisation och Budget.

Instruktioner och mallar hittar du på Naturvårdsverkets [hemsida](#).

Spara ifylld mall som pdf, därefter klicka på mappknappen och leta upp pdf-filen på din dator för att ladda upp filen till ansökningsformuläret.

Se nästa sida för bilaga.

Time plan for projects

Title of the call:

Title of the project proposal:

Personnel involved - insert lines as needed (time is specified in budget)

Project leader Monica Mion (SLU)

Co-worker 1 Max Lindmark (SLU)

Co-worker 2 Karl Lundström (SLU)

Co-worker 4 Karin Hårding (GU)

Co-worker 5 Diet analyst (SLU)

[illegible]

6. Kompetens

The research will be the joint effort of a team of experts from the Swedish University of Agricultural Sciences (SLU) and Göteborg University (GU). The project will be undertaken by Monica Mion (MM), Karl Lundström (KL) and Max Lindmark (ML) at SLU and Professor Karin Hårding (KH) at GU. We are highly qualified to successfully accomplish this multidisciplinary project.

The team has extensive experience of working in both the fisheries and marine mammals' areas and brings together a broad range of expertise which covers the relevant competences required by this project, including:

- Piscivorous diet analyses (KL, MM, KH)
- Fish (MM, ML) and marine mammal (KL, KH) ecology
- Scientific advice on fisheries (ML, MM) and marine mammals (KL, KH)
- Time-series modeling (ML, MM)
- Bioenergetics modeling (ML, KH)
- Support to management agencies (KL, KH)
- Communication with stakeholders (KL, KH, ML, MM)

KL has long-term expertise in ecology of marine mammals and is engaged in various networks and contexts related to research, monitoring, and management of seals. KL has experience from, and is currently involved in, projects on the ecology of seals funded by institutes at both national and EU level. KL supports national and regional authorities in their work on management of seals and is a member of the HELCOM Expert Group on Marine Mammals (EG MaMa) and the ICES Working Group on Marine Mammal Ecology (WGMME).

A major strength of the applicants is a well-balanced blend between analytical expertise and a deep understanding of fish and marine mammal ecology.

Figur

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Personalkostnader

Aktivitetsgrad i projektet

Roll i projektet	Namn	Procent av heltid
1 Projektledare	Monica Mion	50%
2 Medverkande forskare	Max Lindmark	10%
3 Medverkande forskare	Karl Lundström	15%
4 Övrig medverkande	Diet analyst	38%
5 Medverkande forskare	Karin Hårding	3%

Löner inklusive sociala avgifter

Roll i projektet	Namn	Procent av lönen	
1 Projektledare	Monica Mion	50%	
2 Medverkande forskare	Max Lindmark	10%	
3 Medverkande forskare	Karl Lundström	15%	
4 Övrig medverkande	Diet analyst	38%	
Totalt			
2025	2026	2027	Totalt
1 0	408 714	420 975	829 689
2 0	84 697	87 238	171 935
3 0	89 622	184 621	274 243
4 329 878	169 887	0	499 765
Totalt 329 878	752 920	692 834	1 775 632

Övriga kostnader

Driftskostnader

Driftskostnader	Beskrivning	2025	2026	2027	Totalt
1 Travel costs	conference	0	60 000	0	60 000
2 Consultant	data provider	0	150 000	0	150 000
Totalt		0	210 000	0	210 000

Kostnader

Kostnad	2025	2026	2027
Ingen information ifylld			

Total kostnad för projektet

Totalt sökt belopp

Specificerade kostnader	2025	2026	2027	Totalt, sökt	Total kostnad
1 Löner inkl. sociala avgifter	329 878	752 920	692 834	1 775 632	1 775 632
2 Driftskostnader	0	210 000	0	210 000	210 000
3 Kostnad				0	0
4 Delsumma	329 878	962 920	692 834	1 985 632	1 985 632
5 Indirekta kostnader	212 243	484 429	445 770	1 142 442	1 142 442
6 Total projektkostnad	542 121	1 447 349	1 138 604	3 128 074	3 128 074

Annan finansiering

Annan sökt och beviljad finansiering för detta projekt

Finansiär	Sökande/projektledare	Typ av bidrag	Status	Dnr eller motsv.
	2025		2026	2027
Ingen information ifyllt				

CV övriga medverkande

CV övriga medverkande

Ingen fil har laddats upp

CV

CV - Monica Mion

Projektledare: Monica Mion
Födelsedatum: 19880705
Kön: Kvinna
Land: Sverige

Dr-examen: 2021-09-21
Akademisk titel: Doktor
Arbetsgivare: Sveriges lantbruksuniversitet

Doktorsexamen			
Examen	Organisation	Avhandlingens titel (originalspråk)	Handledare
10611. Ekologi, 2021-09-21	Sveriges lantbruksuniversitet	Increasing the biological knowledge of Baltic Sea cod: growth, movements and reproductive potential from historical and contemporary data	Michele Casini

Utbildning

Forskarutbildning			
Examen	Organisation	Avhandlingens titel	Namn på handledare
Doktorsexamen, 10611. Ekologi, 2021-09-21	Sveriges lantbruksuniversitet, Sverige	Increasing the biological knowledge of Baltic Sea cod: growth, movements and reproductive potential from historical and contemporary data	Michele Casini

Utbildning på grund- och avancerad nivå	
År	Examen
2013	10611. Ekologi, Masterexamen, Università di Padova, Italien
2010	10611. Ekologi, Kandidatexamen, Università di Padova, Italien

Arbetsliv

Anställningar

Period	Anställning	Del av forskning i anställningen (%)	Arbetsgivare
juni 2021 - Nuvarande	Environmental analyst specialist, Tillsvideareanställning	50	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser
december 2016 - maj 2021	Doktorand, Meriteringsanställning	100	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser
september 2015 - november 2016	Forskningsassistent, Vikariat	50	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser
januari 2015 - augusti 2015	Lab assistant, Vikariat	20	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser
mars 2014 - december 2014	Consultant, Vikariat	50	ISPRA Istituto Superiore per la Protezione e la Ricerca Ambientale, Italien, Aquatic Resources

CV - Karin Hårding

Medverkande forskare: Karin Hårding
Födelsedatum: 19681001
Kön: Kvinna
Land: Sverige

Dr-examen: 2000-11-17
Akademisk titel: Professor
Arbetsgivare: Göteborgs universitet

Utbildning

Forskarutbildning			
Examen	Organisation	Avhandlingens titel	Namn på handledare
Doktorsexamen, 10608. Zoologi, 2000-11-17	Helsingin yliopisto, Finland, Faculty of Science	Population dynamics of seals, the influence of spatial and temporal structure.	Esa Ranta

Utbildning på grund- och avancerad nivå

År **Examen**

1992 10611. Ekologi, Kandidatexamen, Göteborgs universitet, Sverige

Arbetsliv

Anställningar			
Period	Anställning	Del av forskning i anställningen (%)	Arbetsgivare
oktober 2017 - Nuvarande	Forskarassistent, Tillsvideareanställning	60	Göteborgs universitet, Sverige, Biologi och miljövetenskap, inst för
maj 2021 - Nuvarande	Professor, Tillsvideareanställning	40	Göteborgs universitet, Sverige, Biologi och miljövetenskap, inst för
augusti 2002 - Nuvarande	Lektor, Tillsvideareanställning	59	Göteborgs universitet, Sverige, Biologi och miljövetenskap, inst för

Period	Anställning	Del av forskning i anställningen (%)	Arbetsgivare
april 1995 - april 2001	Forskningsassistent, Projektanställning	100	Naturhistoriska riksmuseet, Sverige, Avdelningen för forskning och samlingar

Postdoktorvistelser		
Period	Organisation	Ämne
juni 2001 - juni 2002	Woods Hole Oceanographic Institution, USA	10699. Annan biologi

Forskarutbyten			
Period	Typ	Organisation	Ämne
januari 2009 - maj 2009	Gästforskare	Bristol University	10611. Ekologi
mars 2007 - juni 2007	Gästforskare	University of Florida	10611. Ekologi

Uppehåll i forskningen	
Period	Beskrivning
2012-11-17 - 2013-11-15	Maternal leave
1996-05-17 - 1997-05-16	Maternal leave
1994-04-29 - 1995-04-21	Maternal leave

Meriter och utmärkelser

Docentur		
År	Ämne	Organisation
2009	106. Biologi (Medicinska tillämpningar under 3 och lantbruksvetenskapliga under 4)	Göteborgs universitet, Sverige, Marina vetenskaper, inst för

Handledda personer			
År	Handledda personer	Lärosäte (handledd person)	Roll
2029	Doktorand, Foteini Kappa		Huvudhandledare
2017	Doktorand, Lisa Sundqvist	Göteborgs universitet, Sverige	Huvudhandledare
2023	Postdok, Batteredine Berhini		Huvudhandledare
2023	Postdok, Darie Carroll		Huvudhandledare
2020	Postdok, Willian Silva		Huvudhandledare

Bidrag erhållna i konkurrens					
Period	Finansiär	Projektledare	Din roll	Delbelopp (kr)	Totalt belopp (kr)
2024 - 2026	Formas, Sverige - Annan forskningsfinansiär	Karin Hårding	Projektledare	0	2 985 340
2022 - 2024	Naturvårdsverket, Sverige - Annan forskningsfinansiär	Karin Hårding	Projektledare	0	2 059 000

Period	Finansiär	Projektledare	Din roll	Delbelopp (kr)	Totalt belopp (kr)
2021 - 2023	Carl Tryggers Stiftelse, Sverige - Övriga privata utförare	Karin Hårding	Projektledare	0	600 000
2021 - 2024	Vetenskapsrådet, Sverige - Universitet och högskolor	Luc Bussiere	Medverkande	0	3 600 000
2021 - 2022	Naturvårdsverket, Sverige - Annan forskningsfinansiär	Karin Hårding	Projektledare	0	840 000

Priser och utmärkelser				
År	Land	Namn på priset/utmärkelsen	Utfärdare	Beskrivning
2008	Sverige	Bengt Wallins Prize	The Royal Society of Arts and Sciences in Gothenburg, Sweden	For promoting research collaboration between Chalmers and Gothenburg University in theoretical population biology.
2002	Sverige	PhD thesis prize from King Carl XVI Gustaf 50th Anniversary Fund for Science, Technology and the Environment	KSLA	Award from for the doctoral thesis on the biology and conservation of marine mammals.
2001	USA	Postdoctoral Fellowship	STINT Research Council in Higher Education	A one year fellowship for a postdoc in Woods Hole Oceanographic. Host Professor Hal Caswell, theoretical population ecology

Andra meriter		
Period	Typ av merit	Beskrivning
2022 - 2022	Bibliometric data	Karin Harding has published 52 peer reviewed scientific articles in journals such as Ecology Letters, American Naturalist, Ecology, Environmental International which have good impact factors. KH has been the main responsible for several research programs, for funding, coordination of data collection, outlined the mathematical models and interpretation of the findings for the majority of these articles. KH has a extensive track record of interdisciplinary collaboration often with world leading experts. Co-authors include scientists in physics, mathematical statistics, computer science and radiophysics in addition to evolutionary ecology and wildlife population dynamics. H-index; 26 Citations: 2 569 Number of peer reviewed book chapters; 2. Conference papers 12. Popular science articles; 5. Reports; 10. Reports with advise for management Naturvårdsverket and HELCOM; 25.
2020 - 2021	Guest Professor	I coordinated a successful application for a guest professor Prof. Scott Edwards (Harvard U. USA) from the Swedish Kings Professorship in Environmental Science. Prof Edwards will visited here during one year, starting in Jan 2021.

Period	Typ av merit	Beskrivning
2018 - 2018	Science Fair "Vetenskapsfestivalen" i Göteborg	I participated in the popular science festival in Gothenburg and gave talks open to the general public on the topic Evolution of Life on earth.. and in space!? together with Professor Cleese Strannegard from Chalmers Technical University.
2013 - 2014	Out reach to school classes	Several lectures to school children, ages 10 to 15 years at Gränna Polar research museum. About my research trip to the Antarctica.
2006 - 2007	Recieved funding for a Tage Erlander guest professor	I coordinated a successful application for a Tage Erlander guest professor. (Prof. John McNamara, Bristol, UK stayed here one year). Funding, the Swedish research council.

CV - Max Lindmark

Medverkande forskare: Max Lindmark
Födelsedatum: 19910315
Kön: Man
Land: Sverige

Dr-examen: 2020-09-15
Akademisk titel: Doktor
Arbetsgivare: Sveriges lantbruksuniversitet

Utbildning

Forskarutbildning			
Examen	Organisation	Avhandlingens titel	Namn på handledare
Doktorsexamen, 10611. Ekologi, 2020-09-15	Sveriges lantbruksuniversitet, Sverige	Temperature- and body size scaling: effects on individuals, populations and food webs	Anna Gårdmark

Utbildning på grund- och avancerad nivå	
År	Examen
2015	10611. Ekologi, Masterexamen, University of Aberdeen, Storbritannien och Nordirland
2014	10611. Ekologi, Kandidatexamen, Göteborgs universitet, Sverige

Arbetsliv

Anställningar				
Period	Anställning	Del av forskning i anställningen (%)	Arbetsgivare	Övrig information
december 2022 - Nuvarande	Forskare, Tillsvidareanställning	100	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser	
juni 2020 - november 2022 (Nuvarande)	Postdoktor, Projektanställning	100	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser	Slutdatum förlängt 6 månader p.g.a. föräldraledighet

Postdoktorvistelser		
Period	Organisation	Ämne
juni 2020 - december 2022	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser	10611. Ekologi

Forskarutbyten			
Period	Typ	Organisation	Ämne
november 2018 - december 2018	Gästforskare	University of Tasmania, Australien, Institute for Marine and Antarctic Studies	10611. Ekologi
mars 2017 - juni 2017	Gästforskare	University of Washington, USA, School of Aquatic And Fishery Sciences	10611. Ekologi

Uppehåll i forskningen	
Period	Beskrivning
2022-01-04 - 2022-07-04	Parental leave

Meriter och utmärkelser

Handledda personer			
År	Handledda personer	Lärosäte (handledd person)	Roll
2024	Doktorand, Henry Hansen		Bihandledare
2024	Postdok, Viktor Thunell		Huvudhandledare
2023	Student, Julia Cao Sanchez		Huvudhandledare
2023	Student, Leo Sheils		Huvudhandledare
2023	Student, Lisa Schüttler		Huvudhandledare

Bidrag erhållna i konkurrens				
Period	Finansiär	Projektledare	Din roll	Totalt belopp (kr)
2024 - 2024	Oscar and Lili Lamm Memorial Foundation, Sverige - Övriga privata utförare	Max Lindmark	Projektledare	995 546
2023 - 2026	Formas, Sverige - Annan forskningsfinansiär	Max Lindmark	Projektledare	3 990 208
2019 - 2019	Sven och Dagmar Saléns stiftelse, Sverige - Övriga privata utförare	Max Lindmark	Projektledare	5 616
2018 - 2018	Knut och Alice Wallenbergs Stiftelse, Sverige - Annan forskningsfinansiär	Max Lindmark	Projektledare	24 000
2016 - 2016	SLU funds for internationalization of graduate education, Sverige - Universitet och högskolor	Max Lindmark	Projektledare	28 000

Priser och utmärkelser				
År	Land	Namn på priset/utmärkelsen	Utfärdare	Beskrivning
2024	USA	SORTEE Open Science Researcher Award (finalist, winner decided by lottery)	SORTEE	

År	Land	Namn på priset/utmärkelsen	Utfärdare	Beskrivning
2015	Storbritannien och Nordirland	Lindsay Laird Prize		In recognition of all-round performance in the Applied Marine and Fisheries Ecology program throughout the year. Awarded jointly with another student
2014	Storbritannien och Nordirland	Fishmongers' Award	Fishmongers' Company	Full fees payment (£3400) awarded to 1 MRes/MSc student on academic merit by the Fishmongers' Company
2014	Sverige	Stipend from Gothenburg Biological Society	Gothenburg Biological Society	For well accomplished bachelor's thesis: By-catch in pelagic fisheries: A study on by-catch in Swedish herring fisheries on the west coast in the winter of 2013/2014

Andra meriter		
Period	Typ av merit	Beskrivning
2017 - 2024	Summary of publications	I have authored or coauthored 13 scientific peer-reviewed publications with 420 citations on Scholar Google and an H-index of 8 (extracted September 6th).
2024 - 2024	Preprints and papers in review	Audzijonyte, A., et al. 2024. Temperature affects fish body sizes. Which sizes? 10.22541/au.171813253.38400021/v1 Ortega-Cisneros, K. et al. 2024. An Integrated Global-to-Regional Scale Workflow for Simulating Climate Change Impacts on Marine Ecosystems. 10.22541/essoar.171587234.44707846/v1 Lindmark, M. et al. 2024. Quantifying competition between two demersal fish species from spatiotemporal stomach content data. https://doi.org/10.1101/2024.04.22.590538 Blanchard, J. L. et al. 2024. Detecting, attributing, and projecting global marine ecosystem and fisheries change: FishMIP 2.0. 10.22541/essoar.170594183.33534487/v1 Lindmark, M. et al. 2024. Stronger effect of temperature on body growth in cool than in warm populations suggests lack of local adaptation. https://doi.org/10.1101/2024.01.17.575983 Maioli, M. et al. 2023. Assessing the overlap between fishing activities and chondrichthyan distribution exposes high-risk areas for bycatch of threatened species. https://doi.org/10.1101/2023.10.25.563919
2022 - 2024	Project leadership	I was a co-lead of the case studies of a project funded by the European Climate, Infrastructure and Environment Executive Agency (CINEA) to work on modelling of fish diet data over time and space. I am the PI of two research grants awarded in competition ()

Period	Typ av merit	Beskrivning
2022 - 2023	Outreach & Science communication	I have ever since I started my undergraduate studies been actively involved in science communication. I was recently interviewed about the paper "Larger but younger fish when growth outpaces mortality in heated ecosystem" on The Naked Scientist podcast. During my PhD, I co-managing research group's Instagram account @fishinfoodwebs (2016–2020), I gave a talk on the effects of warming on fishes at a science evening arranged by the high school in Östhammar municipality. During my undergraduate studies (2014), I gave a popular talk for the Gothenburg Biological Society at the Gothenburg Museum of Natural History on bycatch in small scale pelagic fisheries on the west coast of Sweden. During my involvement in the Swedish Society for Nature Conservation (2011–2014), I gave talks on the topic of toxins in the Baltic herring at local festivals in Gothenburg ("go: TO SEA" and "Västerhavsveckan"), and arranged seminars (4*2 per year) with invited speakers, covering all things marine research.
2021 - 2023	Invited presentations	2021: I was invited to the Gulf of Maine Research Institute to give a May Seminar (GMRI) (by video due to covid), Title: Understanding the effects of climate warming on food webs via individual-level physiology 2022: Invited Speaker to the organization Fiskhälsa (Uppsala), Title: Fish and fisheries in a changing climate 2023: 3rd Internal Water Seminar at SLU (Uppsala), Title: Embracing local scale processes in climate-driven range shifts 2023: Invited to the PICES Annual Science Conference October 2023 Seattle US, Title: Connecting Science and Communities for Sustainable Seas, as an Invited Speaker.

CV - Karl Lundström

Medverkande forskare: Karl Lundström
Födelsedatum: 19710928
Kön: Man
Land: Sverige

Dr-examen: 2012-06-04
Akademisk titel: Doktor
Arbetsgivare: Sveriges lantbruksuniversitet

Utbildning

Forskarutbildning			
Examen	Organisation	Avhandlingens titel	Namn på handledare
Doktorsexamen, 10611. Ekologi, 2012-06-04	Göteborgs universitet, Sverige, Biologi och miljövetenskap, inst för	Assessment of dietary patterns and prey consumption of marine mammals - Grey seals (<i>Halichoerus grypus</i>) in the Baltic Sea	Olle Karlsson

Utbildning på grund- och avancerad nivå

År	Examen
2002	10699. Annan biologi, Magisterexamen, Göteborgs universitet, Sverige
1995	20601. Medicinsk laboratorie- och mätteknik, Högskoleingenjörsexamen/motsv, Lunds universitet, Sverige

Arbetsliv

Anställningar

Period	Anställning	Del av forskning i anställningen (%)	Arbetsgivare	Övrig information
april 2012 - Nuvarande	Forskare, Tillsvidareanställning	25	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser	Researcher
juli 2011 - mars 2012	Biologist, Tillsvidareanställning	10	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser	
mars 2008 - juni 2011	Biologist, Tillsvidareanställning	10	Fiskeriverket	Kustlaboratoriet
mars 2001 - januari 2003	Fishery biologist, Tillsvidareanställning	10	Fiskeriverket	Enheten för resursförvaltning och miljösamordning

Meriter och utmärkelser

Handledda personer			
År	Handledda personer	Lärosäte (handledd person)	Roll
2021	Doktorand, Isa Wallin	Sveriges lantbruksuniversitet, Sverige, Akvatiska resurser	Bihandledare
2020	Doktorand, Malin Tverin	Helsingin yliopisto, Finland, Institutionen för biovetenskaper	Bihandledare
2024	Student, Alexandru Amariei Perez	Uppsala universitet, Sverige, 146 Inst för ekologi och genetik	Huvudhandledare
2024	Student, Jamie Harrigan Branch	Linköpings universitet, Sverige, Institutionen för fysik, kemi och biologi (IFM)	Huvudhandledare
2021	Student, Frederick Fleet	Göteborgs universitet, Sverige, Marina vetenskaper, inst för	Huvudhandledare

Bidrag erhållna i konkurrens					
Period	Finansiär	Projektledare	Din roll	Delbelopp (kr)	Totalt belopp (kr)
2022 - 2024	Formas, Sverige - Annan forskningsfinansiär	Valerio Bartolino	Medverkande	0	2 999 787
2021 - 2024	Nordic Council of Ministers (Nordiska Ministerrådet), Ej Sverige - Internationella organisationer, Sverige	Karl Lundström	Projektledare	0	745 708
2020 - 2022	Europeiska Unionen (EU),	Karl Lundström	Projektledare	0	1 800 000
2020 - 2022	Europeiska Unionen (EU),	Stenungsunds kommun	Medverkande	823 050	1 503 767
2012 - 2013	Europeiska Unionen (EU),	Finnish Game and Fisheries Research Institute	Medverkande	2 500 000	11 000 000

Andra meriter		
Period	Typ av merit	Beskrivning
2018 - 2024	Member of ICES Working Group on Marine Mammal Ecology (WGMME)	Member of the ICES Working Group on Marine Mammal Ecology (WGMME). Responsible for report chapters about impact of seals on fish stocks and ecological role of marine mammals.
2014 - 2024	Member of the HELCOM seal expert group HELCOM SEAL	Member of the HELCOM expert group on marine mammals (EG MAMA).
2011 - 2024	Expertise support on ecology of marine mammals and seabirds to national management authorities	Continuous support with expertise on ecology of marine mammals and seabirds to national management authorities (The Swedish Environmental Protection Agency, The Swedish Agency for Marine and Water Management, County Administrative Boards and The Swedish Board of Agriculture).
2011 - 2024	Communication and interaction with stakeholders	Long-term experience from participation (meetings, workshops, discussions etc.) with different stakeholders on the topic of interactions between seals, cormorants, fish, fisheries and people.

Publikationer

Publikationer - Mion, Monica

Projektledare: Monica Mion
Födelsedatum: 19880705
Kön: Kvinna
Land: Sverige

Dr-examen: 2021-09-21
Akademisk titel: Doktor
Arbetsgivare: Sveriges lantbruksuniversitet

Vetenskaplig publikation - fackgranskade

[Originalartikel i vetenskaplig tidskrift](#)

Titel: Into the wild: coupling otolith and archival tag records to test assumptions underpinning otolith chemistry applications in wild fish

Författare: Karin Hüssy, Stefanie Haase, Monica Mion, Annelie Hilvarsson, Krzysztof Radtke, Tonny B. Thomsen, Maria Krüger-Johnsen, Michele Casini, Anna M. Sturrock

Publiceringsdatum: 2024-03-18 **Volym:** 11

Namn på tidskrift: Frontiers in Marine Science

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Chronological records of elemental concentrations in fish otoliths are a widely used tool to infer the environmental conditions experienced by individual fish. To interpret elemental signals within the otolith, it is important to understand how both external and internal factors impact ion uptake, transport and incorporation. In this study, we have combined chronological records from otoliths and archival data storage tags to quantify the influence of internal (sex, size, age, growth) and external (temperature, depth, salinity) conditions on otolith elemental chemistry of cod (*Gadus morhua*) in natural settings of the Baltic Sea. This study focused on elements primarily under physiological control: Phosphorus (P), magnesium (Mg) and zinc (Zn); and elements under environmental control: Strontium (Sr), barium (Ba) and manganese (Mn). Based on known spatial and temporal patterns in environmental conditions and fish size, growth, and maturity, we posed a series of hypotheses of expected otolith element patterns. Partial effects of internal and external drivers on element concentration were analyzed using a Linear Mixed Model approach with random variables (fish and year). Predicted effects of otolith concentrations of all elements under physiological control (P, Mg, Zn) showed similar trends, with distinct seasonal patterns (lowest concentration in late spring, highest concentrations in winter), and a positive correlation with water temperature, in addition to higher Zn and lower P in spawning individuals. Predicted effects of otolith concentrations of elements expected to be predominantly under environmental control showed the predicted geographic and depth-related trends based on ambient salinity (Ba) and coastal hypoxia (Mn). However, contrary to expectation, Sr was unrelated to salinity. Predicted otolith Ba, Sr and Mn concentrations also exhibited pronounced seasonal patterns that were out of phase with each other but appeared to be partly explained by spawning/feeding migrations. While performing laboratory validation studies for adult fish is typically not possible, these results highlight the importance of assessing local water chemistry and freshwater endmembers in one's study system before otolith elemental chemistry can be reliably used to reconstruct fish habitat use and environmental histories.

Titel: Stock connectivity patterns and indications of sub-stock component structuring of cod in the Sound in the western Baltic Sea

Författare: Regitze B.C. Lundgreen, Anders Nielsen, Mikael van Deurs, Hans Jakob Olesen, Monica Mion, Stefanie Haase, Michele Casini, Uwe Krumme, Karin Hüssy

Publiceringsdatum: 2023-05-01 **Volym:** 261

Namn på tidskrift: Fisheries Research

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Sustainable management of fish stocks requires knowledge of stock structure and connectivity between spawning and feeding habitats. Cod in the Sound in the western Baltic Sea are an example of a stock component with complex connectivity patterns. Currently cod in this area are managed as part of the western Baltic cod stock, while several studies suggest potential connectivity with the neighbouring stock in Kattegat. Here we assess the degree of ecological connectivity of cod in the Sound with the neighbouring areas using historical tagging data from 1957 to 1987, and contemporary growth data from trawl surveys collected between 2007 and 2021. Furthermore, data from cod tagged outside of the Sound between 1960 and 2018 and recaptured in the Sound was used to examine immigration from outside areas. The ecological connectivity between the Sound and Kattegat appeared to be considerable, primarily during the spawning season. Furthermore, cod tagged in the northern Sound were most likely to be recaptured in Kattegat while cod tagged in the southern part of the Sound were mainly recaptured in the Sound. Only 40 out of 16,789 tagged cod released outside of the Sound were eventually recaptured inside the Sound. Overall, these results highlight the need for further examination into the current stock structure of cod in the western Baltic and adjacent areas.

Titel: Examining fish movement in terms of advection and diffusion: a case study of northeastern Atlantic cod

Författare: Regitze B. C. Lundgreen, Anders Nielsen, Maria Krüger-Johnsen, David Righton, Monica Mion, Krzysztof Radtke, Maris Plikshs, Ari J. Leskelä, Jari Raitaniemi, Christopher A. Griffiths, Michele Casini, Uwe Krumme, Karin Hüsey

Publiceringsdatum: 2022-06-16 **Volym:** 691

Namn på tidskrift: Marine Ecology Progress Series

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Advection (directional movement) and diffusion (dispersed movement) were applied for the first time to describe movement patterns in Atlantic cod in the North Sea and Baltic Sea between 1955 and 2020. The advection-diffusion approach provided more detailed estimates of movement that corresponded to previously observed patterns using different analytical techniques. Spatial patterns were evident with greater movement distances in cod from the North Sea and eastern Baltic Sea compared to the western Baltic and Kattegat-Skagerrak. Furthermore, comparative case studies on different ecotypes in the western and eastern Baltic suggested that inshore cod were more resident compared to offshore cod. This preliminary study highlights the usefulness of the advection-diffusion method to describe movements in fish populations, and can be further expanded by incorporating information on environment and mortality and providing information to spatially explicit population models.

Titel: New perspectives on Eastern Baltic cod movement patterns from historical and contemporary tagging data

Författare: Monica Mion, Christopher A. Griffiths, Valerio Bartolino, Stefanie Haase, Annelie Hilvarsson, Karin Hüsey, Maria Krüger-Johnsen, Uwe Krumme, Regitze Benedicte Carlstedt Lundgreen, Johan Lövgren, Kate McQueen, Maris Plikshs, Krzysztof Radtke, Jari Raitaniemi, Michele Casini

Publiceringsdatum: 2022-05-12 **Volym:** 689

Namn på tidskrift: Marine Ecology Progress Series

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Knowledge of the movement patterns and area utilisation of commercially important fish stocks is critical to management. The Eastern Baltic cod *Gadus morhua*, one of the most commercially and ecologically important stocks in the Baltic Sea, is currently one of the most severely impacted fish stocks in Europe. During the last 2 decades, this stock has experienced drastic decreases in population size, distributional range, individual growth and body condition, all of which may have affected the movements between different areas of the Baltic Sea. In this study, we investigated the seasonal movement patterns of Eastern Baltic cod by re-analysing historical tagging data collected by the countries surrounding the Baltic Sea (1955-1988) and compared historical patterns with contemporary data from a recent international tagging experiment (2016-2019). Our re-analyses of historical data showed the presence of different movement behaviours, i.e. resident or seasonally migratory, with larger distances moved by cod released in the northern and central Baltic areas compared to cod released in the southern Baltic areas. Furthermore, trends from the recent tagging experiment indicate a persistent resident strategy in the southern Baltic area. These findings present additional information on general movement patterns and area utilisation of Eastern Baltic cod that could inform future management actions and aid stock recovery.

Titel: Multidecadal changes in fish growth rates estimated from tagging data: A case study from the Eastern Baltic cod (*Gadus morhua*, Gadidae)

Författare: Monica Mion, Stefanie Haase, Jakob Hemmer-Hansen, Annelie Hilvarsson, Karin Hüsey, Maria Krüger-Johnsen, Uwe Krumme, Kate McQueen, Maris Plikshs, Krzysztof Radtke, Franziska Maria Schade, Francesca Vitale, Michele Casini

Publiceringsdatum: 2020-12-22 **Volym:** 22

Namn på tidskrift: Fish and Fisheries

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Long time series of reliable individual growth estimates are crucial for understanding the status of a fish stock and deciding upon appropriate management. Tagging data provide valuable information about fish growth, and are especially useful when age-based growth estimates and stock assessments are compromised by age-determination uncertainties. However, in the literature there is a lack of studies assessing possible changes in growth over time using tagging data. Here, data from tagging experiments performed in the Baltic Sea between 1971 and 2019 were added to those previously analysed for 1955–1970 to build the most extensive tagging dataset available for Eastern Baltic cod (*Gadus morhua*, Gadidae), a threatened stock with severe age-determination problems. Two length-based methods, the GROTAG model (based on the von Bertalanffy growth function) and a Generalized Additive Model, were used to assess for the first time the potential long-term changes in cod growth using age-independent data. Both methods showed strong changes in growth with an increase until the end of the 1980s (8.6–10.6 cm/year for a 40 cm cod depending on the model) followed by a sharp decline. This study also revealed that the current growth of cod is the lowest observed in the past 7 decades (4.3–5.1 cm/year for a 40 cm cod depending on the model), indicating very low productivity. This study provides the first example of the use of tagging data to estimate multidecadal changes in growth rates in wild fish. This methodology can also be applied to other species, especially in those cases where severe age-determination problems exist.

Titel: Regional and stock-specific differences in contemporary growth of Baltic cod revealed through tag-recapture data

Författare: Kate McQueen, Michele Casini, Bodo Dolk, Stefanie Haase, Jakob Hemmer-Hansen, Annelie Hilvarsson, Karin Hüsey, Monica Mion, Thomas Mohr, Krzysztof Radtke, Franziska Maria Schade, Norbert Schultz, Uwe Krumme

Publiceringsdatum: 2020-08-05 **Volym:** 77 **Utfärdandenummer:** 6

Namn på tidskrift: ICES Journal of Marine Science

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The use of growth estimation methods that depend on unreliable age data has previously hindered the quantification of perceived differences in growth rates between the two cod stocks inhabiting the Baltic Sea. Data from cod tagged in different regions of the Baltic Sea during 2007–2019 were combined, and general linear models were fit to investigate inter-regional (defined as area of release) and inter-stock (assigned to a subset of recaptures using genetic and otolith shape analyses) differences in individual growth. An average-sized cod (364 mm) caught in the western Baltic Sea and assigned to the western Baltic cod stock grew at more than double the rate (145 mm year⁻¹) on average than a cod of the same size caught in the eastern Baltic Sea and assigned to the eastern Baltic cod stock (58 mm year⁻¹), highlighting the current poor conditions for the growth of cod in the eastern Baltic Sea. The regional differences in growth rate were more than twice as large (63 mm year⁻¹) as the stock differences (24 mm year⁻¹). Although the relative importance of environmental and genetic factors cannot be fully resolved through this study, these results suggest that environmental experience may contribute to growth differences between Baltic cod stocks.

Titel: Historical growth of Eastern Baltic cod (*Gadus morhua*): Setting a baseline with international tagging data

Författare: Monica Mion, Annelie Hilvarsson, Karin Hüsey, Uwe Krumme, Maria Krüger-Johnsen, Kate McQueen, Mohamed Esha, Roman Motyka, Alessandro Orio, Maris Plikshs, Krzysztof Radtke, Michele Casini

Publiceringsdatum: 2020-03-01 **Volym:** 223 **Utfärdandenummer:** 105442

Namn på tidskrift: Fisheries Research

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Understanding the growth of commercially exploited fish is crucial in fisheries biology and correct estimations of growth and its change over time are paramount for the evaluation of stock status development. Mark-recapture experiments represent a reliable method to estimate growth when age determination based on otolith reading is uncertain, as is the case of the Eastern Baltic cod stock. In this study, historical data (1955–1970) from tagging experiments on Eastern Baltic cod performed by Sweden, Poland, Denmark, Latvia and Germany were digitised and collated for the first time in a unique dataset to estimate historical von Bertalanffy growth function (VBGF) parameters based on fish length increments using GROTAG model. The estimated VBGF parameters were $L_{\infty} = 98.22$ cm and $k = 0.14$ for the period 1955–1964 ($n = 1151$), $L_{\infty} = 123.61$ cm and $k = 0.09$ for 1965–1970 ($n = 2612$), and $L_{\infty} = 125.60$ cm and $k = 0.09$ for the aggregated period ($n = 3763$). A seasonal growth signal was detected for all the periods, with a peak after the spawning season in early autumn. These estimates are the most thorough historical growth baseline now available for the Eastern Baltic cod and can be compared to ongoing and future tagging experiments contributing to the development of stock assessment models for this stock.

Titel: Effect of fish length and nutritional condition on the fecundity of distressed Atlantic cod *Gadus morhua* from the Baltic Sea

Författare: Monica Mion, Anders Thorsen, Francesca Vitale, Jan Dierking, Jens Peter Herrmann, Bastian Huwer, Burkhard von Dewitz, Michele Casini

Publiceringsdatum: 2018-02-26 **Volym:** 92 **Utfärdandenummer:** 4

Namn på tidskrift: Journal of Fish Biology

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The disappearance of larger individuals and the decrease in individual body condition suffered by Atlantic cod *Gadus morhua* in the eastern Baltic during the past two decades can be expected to affect the stock reproductive output. To investigate this, female *G. morhua* were collected during the spawning and pre-spawning period in 2015–2016. The current individual potential fecundity (FP) of eastern Baltic *G. morhua* was estimated and analysed in relation to total length (LT) and indices of nutritional status such as body condition (K) and hepato-somatic index (IH) using generalized linear models. In addition, the current prevalence of atresia and its potential relation to K were investigated. Moreover, a calibration curve to estimate FP from oocyte diameter, based on the autodiometric oocyte counting method, was established for the first time for eastern Baltic *G. morhua* and can be used for future fecundity studies on this stock. The results showed that FP was mainly positively related to fish length, but K and IH also contributed significantly to the variation in FP. The model predicted that fish with $K = 1.2$ have a FP 51% higher than fish of the same LT with $K = 0.8$. The prevalence of fecundity regulation by atresia was 5.8%, but it was found only in fish in the pre-spawning maturity stage and with low K. Temporal changes in biological features such as the length composition and individual body condition of eastern Baltic *G. morhua*, should be accounted for when estimating stock reproductive potential.

Titel: It's elemental, my dear watson: Validating seasonal patterns in otolith chemical chronologies
Författare: Karin Hüssy, Maria Krüger-Johnsen, Tonny Bernt Thomsen, Benjamin Dominguez Heredia, Tomas Næraa, Karin E. Limburg, Yvette Heimbrand, Kate McQueen, Stefanie Haase, Uwe Krumme, Michele Casini, Monica Mion, Krzysztof Radtke
Volym: 78 **Utfärdandenummer:** 5
Namn på tidskrift: Canadian Journal of Fisheries and Aquatic Sciences
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift
Sammanfattning:
 Accurate age data are essential for reliable fish stock assessment. Yet many stocks suffer from inconsistencies in age interpretation. A new approach to obtain age makes use of the chemical composition of otoliths. This study validates the periodicity of recurrent patterns in 25Mg, 31P, 34K, 55Mn, 63Cu, 64Zn, 66Zn, 85Rb, 88Sr, 138Ba, and 208Pb in Baltic cod (*Gadus morhua*) otoliths from tag–recapture and known-age samples. Otolith P concentrations showed the highest consistency in seasonality over the years, with minima co-occurring with otolith winter zones in the known-age otoliths and in late winter – early spring when water temperatures are coldest in tagged cod . The timing of minima differs between stocks, occurring around February in western Baltic cod and 1 month later in eastern Baltic cod; seasonal maxima are also stock-specific, occurring in August and October, respectively. The amplitude in P is larger in faster-growing western compared with eastern Baltic cod. Seasonal patterns with minima in winter – late spring were also evident in Mg and Mn, but less consistent over time and fish size than P. Chronological patterns in P, and to a lesser extent Mg and Mn, may have the potential to supplement traditional age estimation or to guide the visual identification of translucent and opaque otolith patterns used in traditional age estimation.

Titel: Species assignment from seal diet samples using shape analyses in a machine learning framework
Författare: Monica Mion, Florian Berg, Francesco Saltalamacchia, Valerio Bartolino, Johan Lövgren, Mikaela Bergenius Nord, David Gilljam, Martina Blass, Karl Lundström
Namn på tidskrift: ICES Journal of Marine Science
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift
Sammanfattning:
 The identification and quantification of prey ingested is a limiting factor in trophic ecology studies and is fundamental for assessing the impact of a predator on prey populations. Vendace (*Coregonus albula*) and whitefish (*C. lavaretus*) are two congeneric species, which are commonly preyed on by Baltic ringed seals (*Pusa hispida*). The otoliths of these two species are, however, very similar and distinguishing between them in the seal diet using visual inspection has so far been challenging. Here, otolith shape outline analyses were used in combination with machine learning techniques to discriminate between eroded vendace and whitefish otoliths from ringed seal diet samples. An experiment of in vitro digestion of the otoliths was performed to train a machine learning model. Our model is able to self-assign known digested otoliths back to their species of origin with >90% accuracy. Furthermore, 89% (N=690) of the otoliths collected from digestive tract samples could be successfully assigned to species level, i.e. vendace or whitefish. This method is readily applicable for improved understanding of ringed seal feeding habits and predator-prey interactions as well as large scale applications to generate seal-predation matrix inputs for stock assessments of vendace and whitefish. Further development of the machine learning techniques to discriminate between prey species in seal and other piscivorous diets is strongly encouraged.

Publikationer - Karin Hårding

Medverkande forskare: Karin Hårding	Dr-examen: 2000-11-17
Födelsedatum: 19681001	Akademisk titel: Professor
Kön: Kvinna	Arbetsgivare: Göteborgs universitet
Land: Sverige	

Vetenskaplig publikation - fackgranskade

Originalartikel i vetenskaplig tidskrift

Titel: 120-years of ecological monitoring data shows that the risk of overhunting is increased by environmental degradation for an isolated marine mammal population: The Baltic grey seal
Författare: Daire Carroll, Markus Ahola, Anja Carlsson, Martin Skjöld, Karin Hårding
Publiceringsdatum: 2024-03-27
Namn på tidskrift: Journal of Animal Ecology
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Maternal Transfer and Long-Term Population Effects of PCBs in Baltic Grey Seals Using a New Toxicokinetic–Toxicodynamic Population Model.

Författare: K Mauritsson, J.P. Desforges, K.C. Harding

Publiceringsdatum: 2022-10-15

Namn på tidskrift: Arch Environ Contam Toxicol

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Empirical evidence has shown that historical exposure of polychlorinated biphenyls (PCBs) to Baltic grey seals not only severely affected individual fitness, but also population growth rates and most likely caused the retarded recovery rate of the depleted population for decades. We constructed a new model which we term a toxicokinetic–toxicodynamic (TKTD) population model to quantify these effects. The toxicokinetic sub-model describes in detail the bioaccumulation, elimination and vertical transfer from mother to offspring of PCBs and is linked to a toxicodynamic model for estimation of PCB-related damage, hazard and stress impacts on fertility and survival rates. Both sub-models were linked to a Leslie matrix population model to calculate changes in population growth rate and age structure, given different rates of PCB exposure. Toxicodynamic model parameters related to reproductive organ lesions were calibrated using published historical data on observed pregnancy rates in Baltic grey seal females. Compared to empirical data, the TKTD population model described well the age-specific bioaccumulation pattern of PCBs in Baltic grey seals, and thus, the toxicokinetic parameters, deduced from the literature, are believed to be reliable. The model also captured well the general effects of PCBs on historical population growth rates. The model showed that reduced fertility due to increased PCB exposure causes decreased vertical transfer from mother to offspring and in turn increased biomagnification in non-breeding females. The developed TKTD model can be used to perform population viability analyses of Baltic grey seals with multiple stressors, also including by-catches and different hunting regimes. The model can also be extended to other marine mammals and other contaminants by adjustments of model parameters and thus provides a test bed in silico for new substances.

Titel: An automated work-flow for pinniped surveys: A new tool for monitoring population dynamics.

Författare: E Infantes, D Carroll, W.T.A.F. Silva, T Harkonen, S Edwards, K.C. Harding

Publiceringsdatum: 2022-08-15 **Volym:** 10

Namn på tidskrift: Front. Ecol. Evol

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Detecting changes in population trends depends on the accuracy of estimated mean population growth rates and thus the quality of input data. However, monitoring wildlife populations poses economic and logistic challenges especially in complex and remote habitats. Declines in wildlife populations can remain undetected for years unless effective monitoring techniques are developed, guiding appropriate management actions. We developed an automated survey workflow using unmanned aerial vehicles (drones) to quantify the number and size of individual animals, using the well-studied Scandinavian harbour seal (*Phoca vitulina*) as a model species. We compared ground-based counts using telescopes with manual flights, using a zoom photo/video, and pre-programmed flights producing orthomosaic photo maps. We used machine learning to identify and count both pups and older seals and we present a new method for measuring body size automatically. We evaluate the population's reproductive success using drone data, historical counts and predictions from a Leslie matrix population model. The most accurate and time-efficient results were achieved by performing pre-programmed flights where individual seals are identified by machine learning and their body sizes are measured automatically. The accuracy of the machine learning detector was 95–97% and the classification error was 4.6 ± 2.9 for pups and 3.1 ± 2.1 for older seals during good light conditions. There was a clear distinction between the body sizes of pups and older seals during breeding time. We estimated 320 pups in the breeding season 2021 with the drone, which is well beyond the expected number, based on historical data on pup production. The new high quality data from the drone survey confirms earlier indications of a deteriorating reproductive rate in this important harbour seal colony. We show that aerial drones and machine learning are powerful tools for monitoring wildlife in inaccessible areas which can be used to assess annual recruitment and seasonal variations in body condition.

Titel: A unifying framework for estimating generation time in age-structured populations: implications for phylogenetics and conservation biology.

Författare: J Jonasson, T Harkonen, SV Edwards, L Sundqvist, KC Harding

Publiceringsdatum: 2022-04-19

Namn på tidskrift: American Naturalist

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Origin and expansion of the world's most widespread pinniped: range-wide population genomics of the harbour seal (*Phoca vitulina*).

Författare: X. Liu, et. al.

Publiceringsdatum: 2022-01-23

Namn på tidskrift: Molecular Ecology

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The harbour seal (*Phoca vitulina*) is the most widely distributed pinniped, occupying a wide variety of habitats and climatic zones across the Northern Hemisphere. Intriguingly, the harbour seal is also one of the most philopatric seals, raising questions as to how it colonized its current range. To shed light on the origin, remarkable range expansion, population structure and genetic diversity of this species, we used genotyping-by-sequencing to analyse ~13,500 biallelic single nucleotide polymorphisms from 286 individuals sampled from 22 localities across the species' range. Our results point to a Northeast Pacific origin of the harbour seal, colonization of the North Atlantic via the Canadian Arctic, and subsequent stepping-stone range expansions across the North Atlantic from North America to Europe, accompanied by a successive loss of genetic diversity. Our analyses further revealed a deep divergence between modern North Pacific and North Atlantic harbour seals, with finer-scale genetic structure at regional and local scales consistent with strong philopatry. The study provides new insights into the harbour seal's remarkable ability to colonize and adapt to a wide range of habitats. Furthermore, it has implications for current harbour seal subspecies delineations and highlights the need for international and national red lists and management plans to ensure the protection of genetically and demographically isolated populations.

Titel: The Baltic Sea: an ecosystem with multiple stressors.

Författare: R Dietz, et al

Publiceringsdatum: 2021-03-09

Volym: 147

Utfärdandenummer: 6409

Namn på tidskrift: Environment International

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Bioaccumulation of PCBs, OCPs and PBDEs in marine mammals from West Antarctica

Författare: R., M. Lohmann, M Serageldin, E Brault, R Dickhut, K.C. Harding, T Harkonen, O Karlsson, K Lehnert, J Teilmann

Publiceringsdatum: 2021-02-04

Namn på tidskrift: Frontiers in Marine Science

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

To understand the bioaccumulation and food web dynamics of persistent organic pollutants (POPs) as a function of species, age and sex in Antarctic mammals, blubber samples of 3 killer whales (Type C) and 77 pinnipeds (Weddell, Ross and crabeater seals) were collected from the Southern Ocean, Antarctica. They were analyzed for polychlorinated biphenyls (PCBs), organochlorine pesticides (OCPs) and polybrominated diphenyl ethers (PBDEs). ΣDDTs, Σ29PCBs and chlordanes (12 – 4,600, 13 – 1,600, and < 1.5 – 1,700 ng/g lipid, respectively) were the most abundant POPs. Killer whales typically displayed several times greater concentrations of POPs compared to seals, except for PBDEs. PCBs and PBDEs were consistently higher in adult crabeater and Weddell seal males, and in adult female Ross seals than in other sex and age groups reflecting an age accumulation and possible influence of segregated diet, foraging areas, and metabolic transformation rates. POPs concentrations significantly correlated with gene transcription of nuclear receptors involved in detoxification of contaminants and immune relevant cell mediators in the crabeater seals, indicating possible immunotoxic and deleterious health effects. This represents one of the largest studies on POPs in Antarctic marine predators and highlights the complexity of POPs bioaccumulation.

Titel: Risk for overexploiting a seemingly stable seal population: influence of multiple stressors and hunting

Författare: W.T.A.F. Silva, E Bottagisio, T Harkonen, A Galatius, M.T. Olsen, K.C. Harding

Publiceringsdatum: 2021-01-22

Namn på tidskrift: Ecosphere

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Life cycle bioenergetics of the gray seal (*Halichoerus grypus*) in the Baltic Sea: population response to environmental stress

Författare: W.T.A.F. Silva, K.C. Harding, G.M. Marques, C. Sonne, R. Dietz, T. Harkonen, J.P. Desforjes

Publiceringsdatum: 2020-03-08 **Volym:** 145 **Utfärdandenummer:** 106145

Namn på tidskrift: Environmental International

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Multiple stressors and data deficient populations; a comparative life-history approach sheds new light on the extinction risk of the highly vulnerable Baltic harbour porpoises (*Phocoena phocoena*)

Författare: L. Cervin, T. Harkonen, K.C. Harding

Publiceringsdatum: 2020-03-04 **Volym:** 144

Namn på tidskrift: Environment International

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

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Dr-examen: 2020-09-15

Akademisk titel: Doktor

Arbetsgivare: Sveriges lantbruksuniversitet

Vetenskaplig publikation - fackgranskade

Originalartikel i vetenskaplig tidskrift

Titel: Temperature-Dependence Assumptions Drive Projected Responses of Diverse Size-Based Food Webs to Warming

Författare: Jonathan C.P. Reum, Phoebe Woodworth-Jefcoats, Camilla Novaglio, Romain Forestier, Azta Audzijonyte, Anna Gårdmark, Max Lindmark, Julia L. Blanchard

Publiceringsdatum: 2024-03-13 **Volym:** 12 **Utfärdandenummer:** 3

Namn på tidskrift: Earth's Future

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Food web projections are critical for evaluating potential risks to ecosystems and fisheries under global warming. The temperature dependence of biological processes and regional differences in food web structure are two important sources of uncertainty and variation in climate forced projections of fish communities, but we do not know their magnitude or relative contribution. Here we systematically evaluated a range of different assumptions about temperature-dependence on rates, including size-dependent effects, controlling food intake, metabolism, and non-predation mortality in fishes using species-resolved size spectrum food web models that link individual-level physiological processes to population and community dynamics. We simulated the physiological effect of warming in a range of size-structured food web models calibrated to different marine ecosystems and in simplified trait-based models. Higher food intake in warmed conditions increased total fish biomass, catches, and mean body weight, but these effects were offset by the negative effects of warming on metabolism and mortality, which combined resulted in lower total biomasses and catches for most food webs. These effects were enhanced when warming increased metabolic rates more than food intake, and the outcomes were also sensitive to size dependency of temperature responses. Importantly, these general patterns were not uniform across all food webs—individual functional groups and fish species within food webs responded to warming in different ways depending on their position in the food web and its structure. Hence, caution is warranted when generalizing food web or species outcomes to warming because they are mediated by community interactions. Uncertainty related to temperature dependence and ecological interactions will impact food web projections and should be represented in climate change projections.

Titel: Evaluating drivers of spatiotemporal variability in individual condition of a bottom-associated marine fish, Atlantic cod (*Gadus morhua*)

Författare: Max Lindmark, Sean, C. Anderson, Mayya Gogina, Michele Casini

Publiceringsdatum: 2023-07-01 **Volym:** 80 **Utfärdandenummer:** 5

Namn på tidskrift: ICES Journal of Marine Science

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

An organism's body condition describes its mass given its length and is often positively associated with fitness. The condition of Atlantic cod (*Gadus morhua*) in the Baltic Sea has declined dramatically since the early 1990s, possibly due to increased competition for food and hypoxia. However, the effects of biotic and abiotic variables on body condition have not been evaluated at local scales, which is important given spatial heterogeneity. We evaluate changes in distribution, experienced environmental conditions, and individual-level condition of cod in relation to covariates at different spatial scales using geostatistical models with spatial and spatiotemporal random effects. Sprat, *Saduria entomon*, temperature and oxygen were positively associated with condition, and depth was negatively associated. However, the effects of explanatory variables were small—spatial and spatiotemporal latent variables explained 5.7 times more variation than all covariates together (year excluded). Weighting environmental oxygen with local biomass densities revealed steeper declining trends compared to the unweighted oxygen in the environment, while the effect of weighting was less clear for condition. Understanding the drivers of spatiotemporal variation in body condition is critical for predicting responses to environmental change and to effective fishery management; yet low explanatory power of covariates on individual condition constitutes a major challenge.

Titel: Larger but younger fish when growth outpaces mortality in heated ecosystem

Författare: Max Lindmark, Karlsson Malin, Anna Gårdmark

Publiceringsdatum: 2023-05-09 **Volym:** 12

Namn på tidskrift: eLife

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Ectotherms are predicted to “shrink” with global warming, in line with general growth models and the temperature-size rule (TSR), both predicting smaller adult sizes with warming. However, they also predict faster juvenile growth rates and thus larger size-at-age of young organisms. Hence, the result of warming on the size-structure of a population depends on the interplay between how mortality rate, juvenile- and adult growth rates are affected by warming. Here, we use two-decade long time series of biological samples from a unique enclosed bay heated by cooling water from a nearby nuclear power plant to become 5–10 °C warmer than its reference area. We used growth-increment biochronologies (12658 reconstructed length-at-age estimates from 2426 individuals) to quantify how >20 years of warming has affected body growth and size-at-age and catch data to quantify mortality rates and population size- and age structure of Eurasian perch (*Perca fluviatilis*). In the heated area, growth rates were faster for all sizes, and hence size-at-age was larger for all ages, compared to the reference area. While mortality rates were also higher (lowering mean age by 0.4 years), the faster growth rates lead to a 2 cm larger mean size in the heated area. Differences in the size-spectrum exponent (describing how the abundance varies with size) were less clear statistically. Our analyses reveal that mortality, in addition to plastic growth and size-responses, is a key factor determining the size structure of populations exposed to warming. Understanding the mechanisms by which warming affects the size- and age structure of populations is critical for predicting the impacts of climate change on ecological functions, interactions, and dynamics.

Titel: Mechanistic temperature-size rule explanation should reconcile physiological and mortality responses to temperature

Författare: Asta Audzijonyte, Egle Jakubavičiūtė, Max Lindmark, Shane. A Richards

Publiceringsdatum: 2022-10-18 **Volym:** 243 **Utfärdandenummer:** 2

Namn på tidskrift: The Biological Bulletin

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The temperature-size rule is one of the universal rules in ecology and states that ectotherms in warmer waters will grow faster as juveniles, mature at smaller sizes and younger ages, and reach smaller maximum body sizes. Many models have unsuccessfully attempted to reproduce temperature-size rule-consistent life histories by using two-term (anabolism and catabolism) Pütter-type growth models, such as the von Bertalanffy. Here, we present a physiologically structured individual growth model, which incorporates an energy budget and optimizes energy allocation to growth, reproduction, and reserves. Growth, maturation, and reproductive output emerge as a result of life-history optimization to specific physiological rates and mortality conditions. To assess which processes can lead to temperature-size rule-type life histories, we simulate 42 scenarios that differ in temperature and body size dependencies of intake, metabolism, and mortality rates. Results show that the temperature-size rule can emerge in two ways. The first way requires both intake and metabolism to increase with temperature, but the temperature-body size interaction of the two rates must lead to relatively faster intake increase in small individuals and relatively larger metabolism increase in large ones. The second way requires only higher temperature-driven natural mortality and faster intake rates in early life (no change in metabolic rates is needed). This selects for faster life histories with earlier maturation and increased reproductive output. Our model provides a novel mechanistic and evolutionary framework for identifying the conditions necessary for the temperature-size rule. It shows that the temperature-size rule is likely to reflect both physiological changes and life-history optimization and that use of von Bertalanffy-type models, which do not include reproduction processes, can hinder our ability to understand and predict ectotherm responses to climate change.

Titel: Temperature impacts on fish physiology and resource abundance lead to faster growth but smaller fish sizes and yields under warming

Författare: Max Lindmark, Asta Audzijonyte, Julia L. Blanchard, Anna Gårdmark

Publiceringsdatum: 2022-07-13 **Volym:** 28 **Utfärdandenummer:** 21

Namn på tidskrift: Global Change Biology

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Resolving the combined effect of climate warming and exploitation in a food web context is key for predicting future biomass production, size-structure and potential yields of marine fishes. Previous studies based on mechanistic size-based food web models have found that bottom-up processes are important drivers of size-structure and fisheries yield in changing climates. However, we know less about the joint effects of 'bottom-up' and physiological effects of temperature; how do temperature effects propagate from individual-level physiology through food webs and alter the size-structure of exploited species in a community? Here, we assess how a species-resolved size-based food web is affected by warming through both these pathways and by exploitation. We parameterize a dynamic size spectrum food web model inspired by the offshore Baltic Sea food web, and investigate how individual growth rates, size-structure, and relative abundances of species and yields are affected by warming. The magnitude of warming is based on projections by the regional coupled model system RCA4-NEMO and the RCP 8.5 emission scenario, and we evaluate different scenarios of temperature dependence on fish physiology and resource productivity. When accounting for temperature-effects on physiology in addition to on basal productivity, projected size-at-age in 2050 increases on average for all fish species, mainly for young fish, compared to scenarios without warming. In contrast, size-at-age decreases when temperature affects resource dynamics only, and the decline is largest for young fish. Faster growth rates due to warming, however, do not always translate to larger yields, as lower resource carrying capacities with increasing temperature tend to result in decline in the abundance of larger fish and hence spawning stock biomass. These results suggest that to understand how global warming affects the size structure of fish communities, both direct metabolic effects and indirect effects of temperature via basal resources must be accounted for.

Titel: Optimum growth temperature declines with body size within fish species

Författare: Max Lindmark, Jan Ohlberger, Anna Gårdmark

Publiceringsdatum: 2022-01-21 **Volym:** 28 **Utfärdandenummer:** 7

Namn på tidskrift: Global Change Biology

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

According to the temperature-size rule, warming of aquatic ecosystems is generally predicted to increase individual growth rates but reduce asymptotic body sizes of ectotherms. However, we lack a comprehensive understanding of how growth and key processes affecting it, such as consumption and metabolism, depend on both temperature and body mass within species. This limits our ability to inform growth models, link experimental data to observed growth patterns, and advance mechanistic food web models. To examine the combined effects of body size and temperature on individual growth, as well as the link between maximum consumption, metabolism, and body growth, we conducted a systematic review and compiled experimental data on fishes from 52 studies that combined body mass and temperature treatments. By fitting hierarchical models accounting for variation between species, we estimated how maximum consumption and metabolic rate scale jointly with temperature and body mass within species. We found that whole-organism maximum consumption increases more slowly with body mass than metabolism, and is unimodal over the full temperature range, which leads to the prediction that optimum growth temperatures decline with body size. Using an independent dataset, we confirmed this negative relationship between optimum growth temperature and body size. Small individuals of a given population may, therefore, exhibit increased growth with initial warming, whereas larger conspecifics could be the first to experience negative impacts of warming on growth. These findings help advance mechanistic models of individual growth and food web dynamics and improve our understanding of how climate warming affects the growth and size structure of aquatic ectotherms.

Titel: Effects of warming on intraguild predator communities with ontogenetic diet-shifts

Författare: Viktor Thunell, Max Lindmark, Magnus Huss, Anna Gårdmark

Publiceringsdatum: 2021-10-11 **Volym:** 198 **Utfärdandenummer:** 6

Namn på tidskrift: The American Naturalist

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Species interactions mediate how warming affects community composition via individual growth and population size structure. While predictions on how warming affects composition of size or stage-structured communities have so far focused on linear (food chain) communities, mixed competition-predation interactions, such as intraguild predation, are common. Intraguild predation often results from changes in diet over ontogeny ("ontogenetic diet shifts") and strongly affects community composition and dynamics. Here, we study how warming affects a community of intraguild predators with ontogenetic diet shifts, consumers, and shared prey by analyzing a stage-structured bioenergetics multispecies model with temperature and body size-dependent individual-level rates. We find that warming can strengthen competition and decrease predation, leading to a loss of a cultivation mechanism (the feedback between predation on and competition with consumers exerted by predators) and ultimately predator collapse. Furthermore, we show that the effect of warming on community composition depends on the extent of the ontogenetic diet shift and that warming can cause a sequence of community reconfigurations in species with partial diet shifts. Our findings contrast previous predictions concerning individual growth of predators and the mechanisms behind predator loss in warmer environments and highlight how feedbacks between temperature and intraspecific size structure are important for understanding such effects on community composition.

Titel: Experimental evidence of gradual size-dependent shifts in body size and growth of fish in response to warming

Författare: Magnus Huss, Max Lindmark, Philip Jacobson, Renee. M van Dorst, Anna Gårdmark

Publiceringsdatum: 2019-04-01 **Volym:** 25 **Utfärdandenummer:** 7

Namn på tidskrift: Global Change Biology

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

A challenge facing ecologists trying to predict responses to climate change is the few recent analogous conditions to use for comparison. For example, negative relationships between ectotherm body size and temperature are common both across natural thermal gradients and in small-scale experiments. However, it is unknown if short-term body size responses are representative of long-term responses. Moreover, to understand population responses to warming, we must recognize that individual responses to temperature may vary over ontogeny. To enable predictions of how climate warming may affect natural populations, we therefore ask how body size and growth may shift in response to increased temperature over life history, and whether short- and long-term growth responses differ. We addressed these questions using a unique setup with multidecadal artificial heating of an enclosed coastal bay in the Baltic Sea and an adjacent reference area (both with unexploited populations), using before-after control-impact paired time-series analyses. We assembled individual growth trajectories of ~13,000 unique individuals of Eurasian perch and found that body growth increased substantially after warming, but the extent depended on body size: Only among small-bodied perch did growth increase with temperature. Moreover, the strength of this response gradually increased over the 24 year warming period. Our study offers a unique example of how warming can affect fish populations over multiple generations, resulting in gradual changes in body growth, varying as organisms develop. Although increased juvenile growth rates are in line with predictions of the temperature–size rule, the fact that a larger body size at age was maintained over life history contrasts to that same rule. Because the artificially heated area is a contemporary system mimicking a warmer sea, our findings can aid predictions of fish responses to further warming, taking into account that growth responses may vary both over an individual's life history and over time.

Titel: Size-based ecological interactions drive food web responses to climate warming

Författare: Max Lindmark, Jan Ohlberger, Magnus Huss, Anna Gårdmark

Publiceringsdatum: 2019-02-28 **Volym:** 22 **Utfärdandenummer:** 5

Namn på tidskrift: Ecology Letters

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Predicting climate change impacts on animal communities requires knowledge of how physiological effects are mediated by ecological interactions. Food-dependent growth and within-species size variation depend on temperature and affect community dynamics through feedbacks between individual performance and population size structure. Still, we know little about how warming affects these feedbacks. Using a dynamic stage-structured biomass model with food-, size- and temperature-dependent life history processes, we analyse how temperature affects coexistence, stability and size structure in a tri-trophic food chain, and find that warming effects on community stability depend on ecological interactions. Predator biomass densities generally decline with warming – gradually or through collapses – depending on which consumer life stage predators feed on. Collapses occur when warming induces alternative stable states via Allee effects. This suggests that predator persistence in warmer climates may be lower than previously acknowledged and that effects of warming on food web stability largely depend on species interactions.

Titel: Temperature-dependent body size effects determine population responses to climate warming

Författare: Max Lindmark, Magnus Huss, Jan Ohlberger, Anna Gårdmark

Publiceringsdatum: 2017-11-21 **Volym:** 21 **Utfärdandenummer:** 2

Namn på tidskrift: Ecology Letters

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Current understanding of animal population responses to rising temperatures is based on the assumption that biological rates such as metabolism, which governs fundamental ecological processes, scale independently with body size and temperature, despite empirical evidence for interactive effects. Here, we investigate the consequences of interactive temperature- and size scaling of vital rates for the dynamics of populations experiencing warming using a stage-structured consumer-resource model. We show that interactive scaling alters population and stage-specific responses to rising temperatures, such that warming can induce shifts in population regulation and stage-structure, influence community structure and govern population responses to mortality. Analysing experimental data for 20 fish species, we found size-temperature interactions in intraspecific scaling of metabolic rate to be common. Given the evidence for size-temperature interactions and the ubiquity of size structure in animal populations, we argue that accounting for size-specific temperature effects is pivotal for understanding how warming affects animal populations and communities.

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Vetenskaplig publikation - fackgranskade

Originalartikel i vetenskaplig tidskrift

Titel: Report of the ICES Working Group on Marine Mammal Ecology (WGMME)

Författare: ICES ,

Publiceringsdatum: 2024-01-25 **Volym:** 5 **Utfärdandenummer:** 88

Namn på tidskrift: ICES Scientific Reports

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Five terms of reference (ToRs) were addressed at the working group. The first three terms of reference were standing ones. Under ToR A, new information on cetacean and seal population abundance, distribution, and population/stock structure, was reviewed, including information on vagrancy in cetacean and pinniped species. For cetaceans, coverage from the latest SCANS-IV survey (summer 2022) was presented as well as the results of recent regional/national surveys, particularly those in the Bay of Biscay and around the Iberian Peninsula. Updates on population estimates and distribution were provided for particular species studies, such as some coastal bottlenose dolphin populations. For seals, latest monitoring results were given for harbour, grey, and Baltic and Saimaa ringed seals. In addition, where possible, local long-term trends were illustrated for those species, based on earlier efforts by WGMME to assemble these data into a seal database. For both species' groups, recent records of vagrant species were summarised. Under ToR B, cetacean and seal management frameworks in the North Atlantic were discussed, with an overview of the EU Maritime Spatial Planning Directive, and examples from the United Kingdom, Spain and the Faroe Islands of national management frameworks regarding marine mammals. ToR C provided an overview of new published information with regards to anthropogenic threats to marine mammal populations following on from the review by WGMME in 2015 (ICES, 2015) and subsequent updates. These were considered under the following headings: cumulative effects, fishery interactions, chemical pollution including marine debris, underwater noise, ship strikes and other physical trauma, tourism disturbance, climate change, and new pathogens (including avian influenza). ToR D focused upon bycatch. In support of WGBYC, this ToR aimed to contribute to the Roadmap for ICES PETS bycatch advice. ToR E involved liaison with other WGs. The Chairs of the newly-formed WGJCDP introduced to WGMME members, the Joint Cetacean Database Programme, which is to be hosted by the ICES Data Centre. The scope to collect information on other marine species besides cetaceans was discussed. A meeting with another newly formed ICES working group, on Marine Protected Areas, was planned but was deferred at the request of that group. On behalf of the working group, the Chairs would like to thank The Swedish Museum of Natural History for hosting the meeting.

Titel: Detection of multiple fish species in the diet of the invasive round goby reveals new trophic interactions in the Baltic Sea

Författare: Isa Wallin Kihlberg, Ann-Britt Florin, Karl Lundström, Örjan Östman

Publiceringsdatum: 2023-06-28 **Volym:** 18 **Utfärdandenummer:** 2

Namn på tidskrift: Aquatic Invasions

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The mesopredatory round goby (*Neogobius melanostomus*) is an important fish invader in fresh and brackish waters of the northern hemisphere. Trophic interactions of invasive species can generate ecological impacts across the food web in invaded ecosystems. Here we investigated major diet components, spatiotemporal variation in diet and the effect of round goby densities on diet composition in two geographically distinct round goby populations in the Baltic Sea. The round goby is a generalist feeder but previous diet studies, based on visual prey identification, have likely over-emphasized the importance of hard-shelled, invertebrate prey in round goby diet, as shells degrade and evacuate slowly relative to soft-bodied prey that break down rapidly in the stomach. We therefore, in addition to visual stomach content analysis, used DNA metabarcoding, which is less biased towards hard body structures of prey and can be used for species assignment of highly degraded prey. The results demonstrated that round goby diet composition varied between areas and years. Visual stomach content analysis indicated that blue mussel was the main prey in the southern area, whereas hydrobiid gastropods were the major diet component in the northern area. Metabarcoding revealed that several fish species, likely the egg or larval stages of e.g. stickleback, cod and herring, were also part of the round goby diet. Analyses suggested that round goby feeding on fishes was positively associated with round goby densities. Our study shows that round goby, in addition to benthic invertebrates, preys on several fish species of ecological and commercial importance. Thus, there is potential for predator-prey reversal and negative effects of the invasive round goby on large, predatory fishes.

Titel: Environmental abundances of the non-native round goby *Neogobius melanostomus* influence feeding of native fish predators

Författare: Hedi Herlevi, Isa Wallin Kihlberg, Katri Aarnio, Erik Bonsdorff, Ann-Britt Florin, Andreas Ljung, Karl Lundström, Johanna Mattila, Örjan Östman

Publiceringsdatum: 2023-03-16 **Volym:** 102 **Utfärdandenummer:** 6

Namn på tidskrift: Journal of Fish Biology

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The authors assessed the importance of the round goby *Neogobius melanostomus* as prey for three native predatory fish species, Atlantic cod *Gadus morhua*, European perch *Perca fluviatilis* and northern pike *Esox lucius*, in a northern and southern area of the Baltic Proper, using a combination of visual analysis and DNA metabarcoding of predator stomach contents. To explore the influence of environmental abundances of *N. melanostomus* on predation, they related the occurrence of *N. melanostomus* in predator diets to its abundance in survey fishing. *Gadus morhua* and *E. lucius* in the southern area showed the highest tendency to feed on *N. melanostomus* when it was abundant, as *N. melanostomus* occurred in up to 100% of stomachs and constituted up to 88% of the total diet volume proportion. The diet contribution of *N. melanostomus* was associated with *N. melanostomus* abundances for *G. morhua* and *E. lucius*, and when *N. melanostomus* was abundant, these predators exhibited lower prey richness and a higher degree of piscivory. *G. morhua* and *P. fluviatilis* also fed less on crustacean prey when *N. melanostomus* was abundant. The high importance of *N. melanostomus* in diets of native fish predators may modify indirect interactions between *N. melanostomus* and native prey species in invaded coastal communities.

Titel: Complementary methods assessing short and long-term prey of a marine top predator – Application to the grey seal-fishery conflict in the Baltic Sea

Författare: Malin Tverin, Rodrigo Esparza-Salas, Annika Strömberg, Patrik Tang, Iiris Kokkonen, Annika Herrero, Kaarina Kauhala, Olle Karlsson, Raisa Tiilikainen, Markus Vetemaa, Tuula Sinisalo, Reijo Käkelä, Karl Lundström

Publiceringsdatum: 2019-01-02 **Volym:** 14 **Utfärdandenummer:** 1

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Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

The growing grey seal (*Halichoerus grypus*) population in the Baltic Sea has created conflicts with local fisheries, comparable to similar emerging problems worldwide. Adequate information on the foraging habits is a requirement for responsible management of the seal population. We investigated the applicability of available dietary assessment methods by comparing morphological analysis and DNA metabarcoding of gut contents (short-term diet; n = 129/125 seals, respectively), and tissue chemical markers i.e. fatty acid (FA) profiles of blubber and stable isotopes (SIs) of liver and muscle (mid- or long-term diet; n = 108 seals for the FA and SI markers). The methods provided complementary information. Short-term methods indicated prey species and revealed dietary differences between age groups and areas but for limited time period. In the central Baltic, herring was the main prey, while in the Gulf of Finland percid and cyprinid species together comprised the largest part of the diet. Perch was also an important prey in the western Baltic Proper. The DNA analysis provided firm identification of many prey species, which were neglected or identified only at species group level by morphological analysis. Liver SIs distinguished spatial foraging patterns and identified potentially migrated individuals, whereas blubber FAs distinguished individuals frequently utilizing certain types of prey. Tissue chemical markers of adult males suggested specialized feeding to certain areas and prey, which suggest that these individuals are especially prone to cause economic losses for fisheries. We recommend combined analyses of gut contents and tissue chemical markers as dietary monitoring methodology of aquatic top predators to support an optimal ecosystem-based management.

Titel: The necessity of a holistic approach when managing marine mammal–fisheries interactions: Environment and fisheries impact are stronger than seal predation

Författare: David Costalago, Barbara Bauer, Maciej T. Tomczak, Karl Lundström, Monika Winder

Publiceringsdatum: 2018-12-08

Namn på tidskrift: Ambio

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

Seal populations are recovering in many regions around the world and, consequently, they are increasingly interacting with fisheries. We used an Ecopath with Ecosim model for the offshore Central Baltic Sea to investigate the interactions between the changes in fish stocks and grey seal (*Halichoerus grypus*) population under different fishing and environmental scenarios for the twenty-first century. The assumed climate, eutrophication and cod (*Gadus morhua*) fisheries scenarios modified seal predation impacts on fish. Fish biomass and catches are more affected by fishing mortality and the environment than by seal predation. Our results highlight that the impacts of the increasing seal population on lower trophic levels are complex; thus, we emphasize the need to consider a range of possible ecosystem contexts when evaluating potential impacts of top predators. Finally, we suggest that an increasing seal population is not likely to hinder the preservation of the main Baltic fish stocks.

Titel: Diet of seals in the Baltic Sea region: a synthesis of published and new data from 1968 to 2013
Författare: Camilla Hjorth Scharff-Olsen, Anders Galatius, Jonas Teilmann, Rune Dietz, Signe May Andersen, Simon Jarnit, Anne-Mette Kroner, Amanda Bolt Botnen, Karl Lundström, Peter Rask Møller, Morten Tange Olsen
Publiceringsdatum: 2018-11-15 **Volym:** 76 **Utfärdandenummer:** 1
Namn på tidskrift: ICES Journal of Marine Science

Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Sammanfattning:

A crucial first step in assessing and managing the role and impacts of predators on their environment is knowledge on their foraging behaviour and diet. Here, we synthesize previously published and newly generated data on the diet of harbour seals, grey seals, and ringed seals in the Baltic Sea region. More than 45 000 otoliths recovered from 3147 samples of scats and digestive tracts were collected throughout all seasons and most Baltic Sea sub-basins from 1968 to 2013. The data revealed a large extent of interspecific, spatial, and seasonal variation in seal diet, implying that caution should be made when extrapolating from one species, area or season, to others. Still, a few fish species, including Atlantic herring, sprat, cod, and sandeels had high occurrence across seal species and locations. The compiled data provide the first overview of seal diet across the entire Baltic Sea region, but also comes with several limitations. Thus, while the data presented here constitutes an important reference for future inference, it also illustrates an urgent need for standardizing methodology across studies on the diet of seals and other aquatic predators.

Titel: Competition for the fish – fish extraction from the Baltic Sea by humans, aquatic mammals, and birds
Författare: Sture Hansson, Ulf Bergström, Erik Bonsdorff, Tero Härkönen, Niels Jepsen, Lena Kautsky, Karl Lundström, Sven-Gunnar Lunneryd, Maria Ovegård, Juhani Salmi, Dmitry Sendek, Markus Vetemaa
Publiceringsdatum: 2017-11-13
Namn på tidskrift: ICES Journal of Marine Science
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: A continuous involvement of stakeholders promotes the ecosystem approach to fisheries in the 8-fjords area on the Swedish west coast
Författare: Andreas Bryhn, Karl Lundström, Amelie Johansson, Henrik Ragnarsson Stabo, Henrik Svedäng
Publiceringsdatum: 2016-12-23 **Volym:** 74 **Utfärdandenummer:** 1
Namn på tidskrift: Ices Journal of Marine Science
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Does predation by grey seals (*Halichoerus grypus*) affect Bothnian Sea herring stock estimates?
Författare: Anna Gårdmark, Örjan Östman, Anders Nielsen, Karl Lundström, Olle Karlsson, Jukka Pönni, Teija Aho
Publiceringsdatum: 2012-06-08 **Volym:** 69 **Utfärdandenummer:** 8
Namn på tidskrift: Ices Journal of Marine Science
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

Titel: Understanding the diet composition of marine mammals: grey seals (*Halichoerus grypus*) in the Baltic Sea
Författare: Karl Lundström, Olle Hjerne, Sven-Gunnar Lunneryd, Olle Karlsson
Publiceringsdatum: 2010-03-30 **Volym:** 67 **Utfärdandenummer:** 6
Namn på tidskrift: Ices Journal of Marine Science
Vetenskaplig publikation - fackgranskade: Originalartikel i vetenskaplig tidskrift

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